

Chapter 3:

Elicited Behaviors & Classical Conditioning

Elicited Behaviors

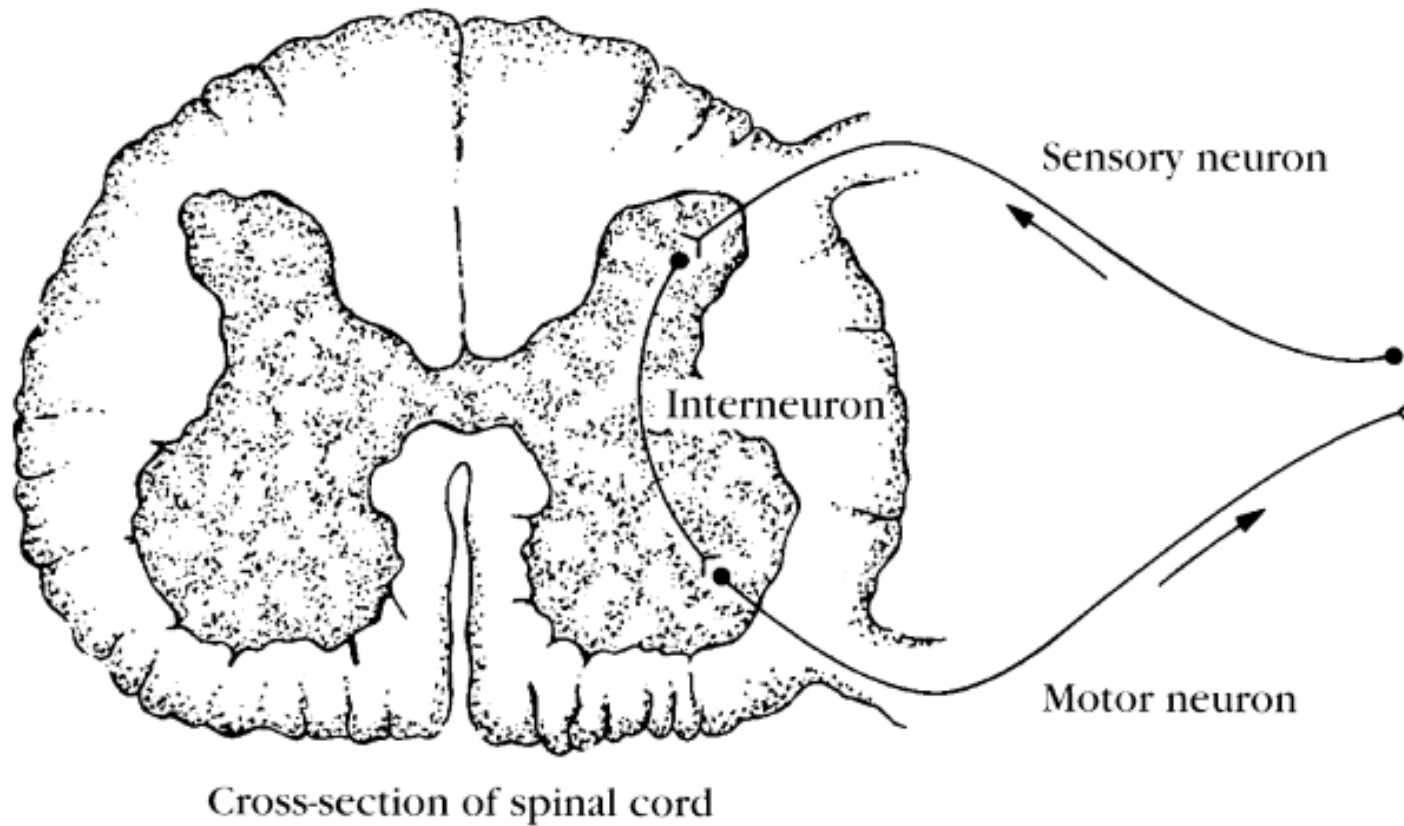
- a behavior that is “drawn out” of an organism by the presentation of a stimulus
- a behavior that is automatic and involuntary



Reflexes

- Basic form of elicited behavior
- Defined as “a relatively simple, automatic response to a stimulus”
- Reflexes are closely tied to survival.
- Examples: *startle response, orienting response, flexion response*

Reflex Arc



Fixed Action Pattern

- Fixed sequence of responses elicited by a sign stimulus (or releaser)

Mechanisms of Learning

- **Habituation & sensitization** tend to be seen across species.
- These processes probably have tremendous survival advantages.
- They help us sort information in our environment into two basic categories: currently relevant and currently irrelevant

Habituation

- We tend to habituate to stimuli that are *currently irrelevant* and *low-intensity*.
- Defined as “a decrease in the strength of an elicited behavior following repeated presentations of the eliciting stimulus”
- Examples:
 - ticking of a clock, noise of traffic, consistent drone of music from neighbor’s house

Dishabituation

- When habituated responses reappear following the presentation of a seemingly irrelevant novel stimulus



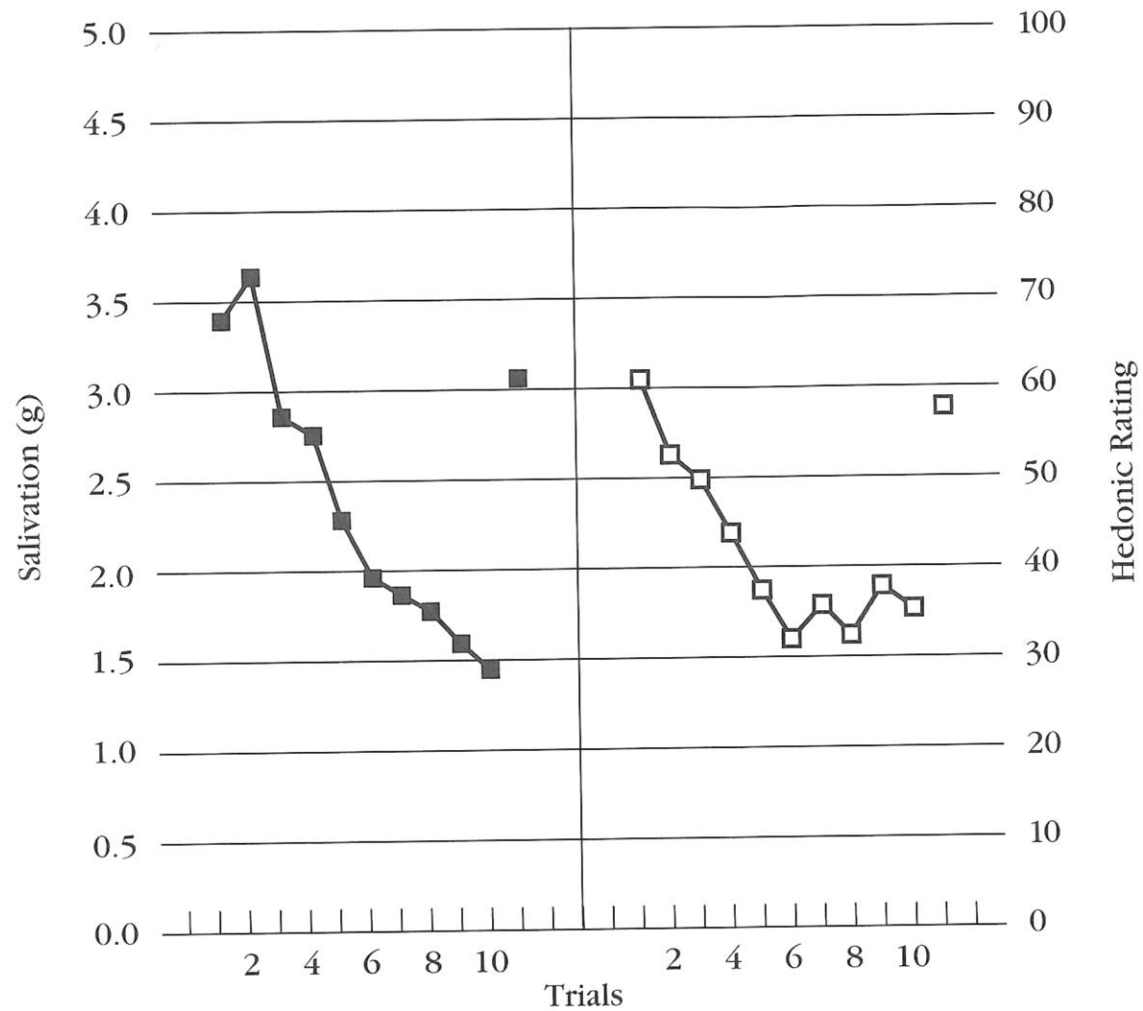


FIGURE 2.5

Salivation and ratings of pleasantness in response to a taste stimulus (lime or lemon) repeatedly presented to women on Trials 1–10. The alternate taste was presented on Trial 11, causing a substantial recovery in responding. (After Epstein, Rodefer, Wisniewski & Caggiula, 1992).

Sensitization

- Defined as “an increase in the strength of an elicited behavior following repeated presentations of the eliciting stimulus”
- Unlike Habituation, Sensitization often generalizes to other stimuli.
- Examples: war veterans’ responses to exploding artillery shells and subsequent startling sounds

Aplysia Californica

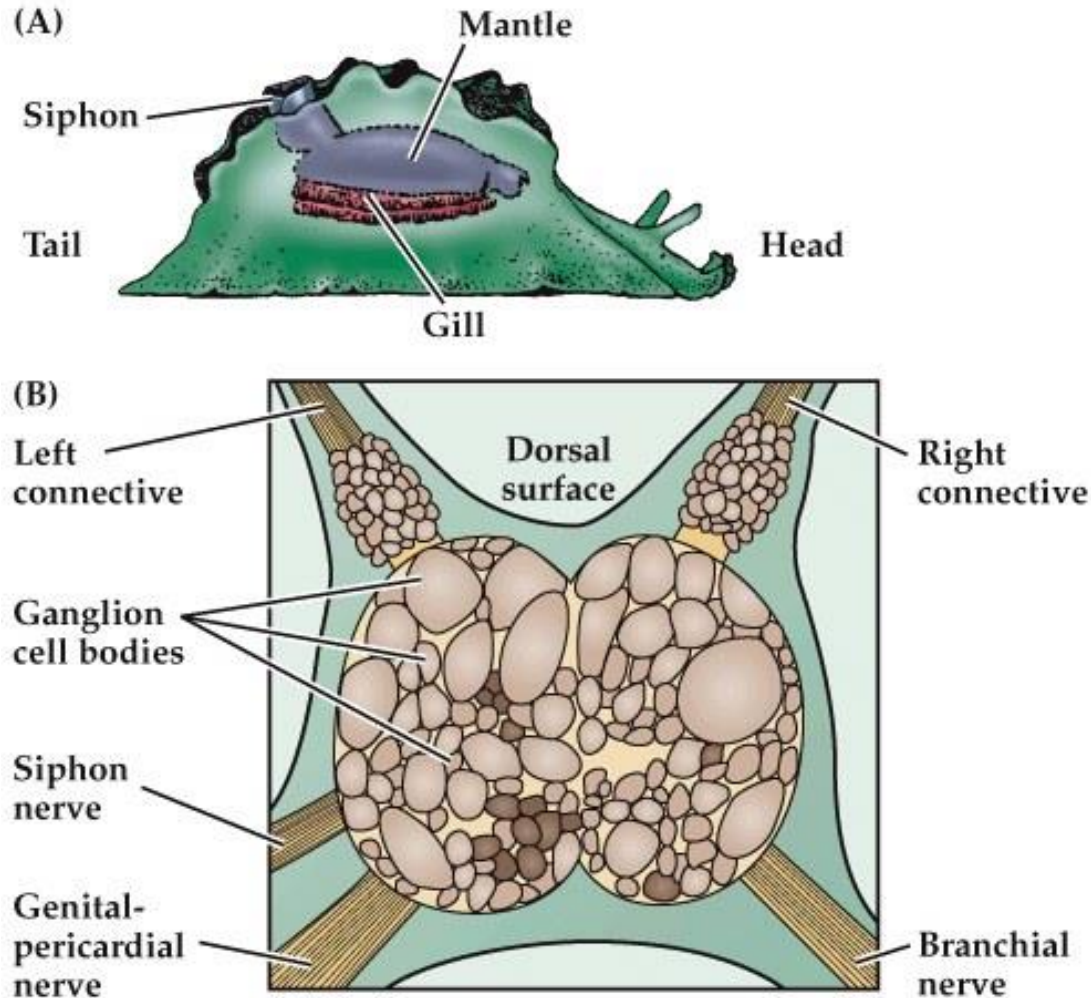
- Long-term synaptic plasticity
- Only has tens of thousands of neurons
- Some large (1mm)
- Stereotyped locations within the ganglia that make up the nervous system
- Aplysia and memory

NOVA

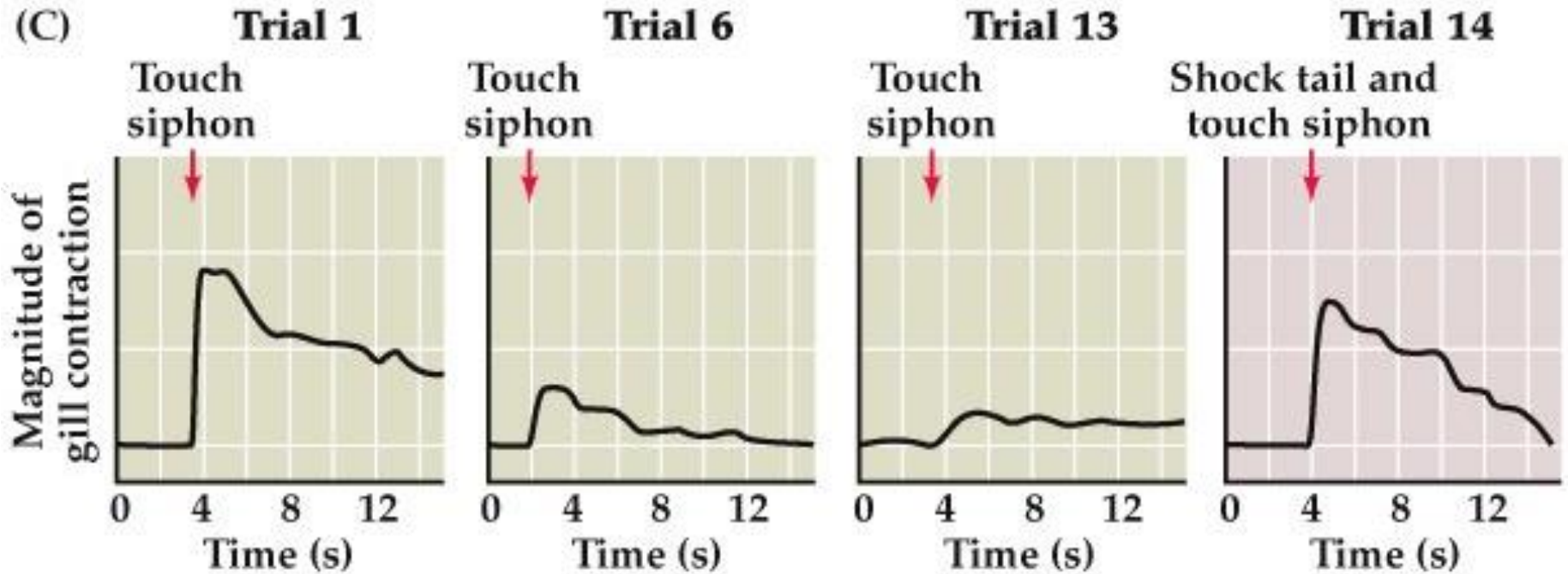
Non-associative Memory

- **Habituation**-process where the animal becomes less responsive to repeated occurrences of a stimulus
- **Sensitization**-allows the animal to generalize an aversive response elicited by a noxious stimulus to a variety of other non-noxious stimuli

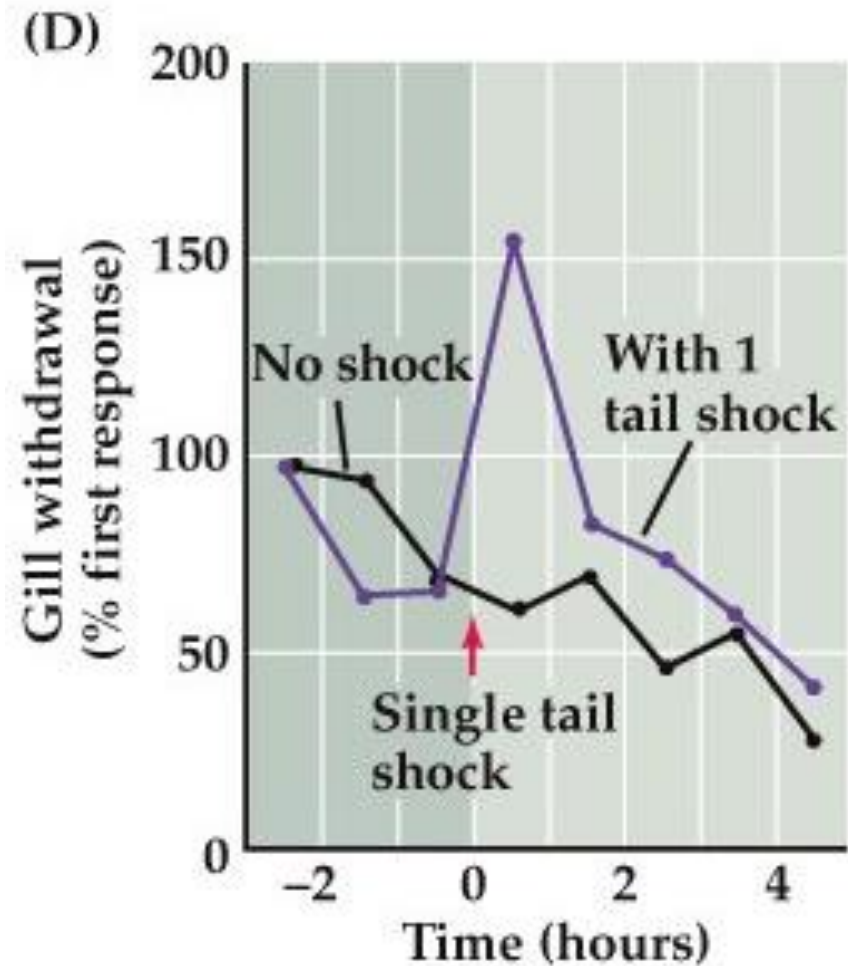
Short-term sensitization of the *Aplysia* gill-withdrawal reflex



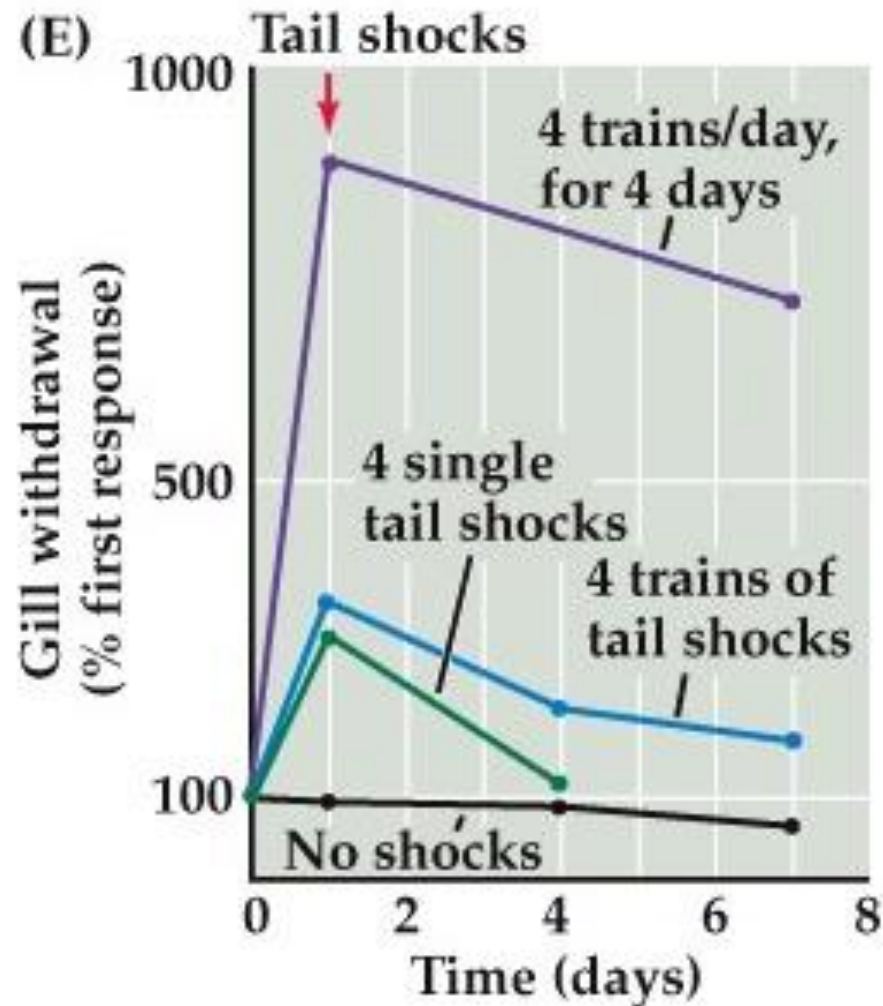
Habituation



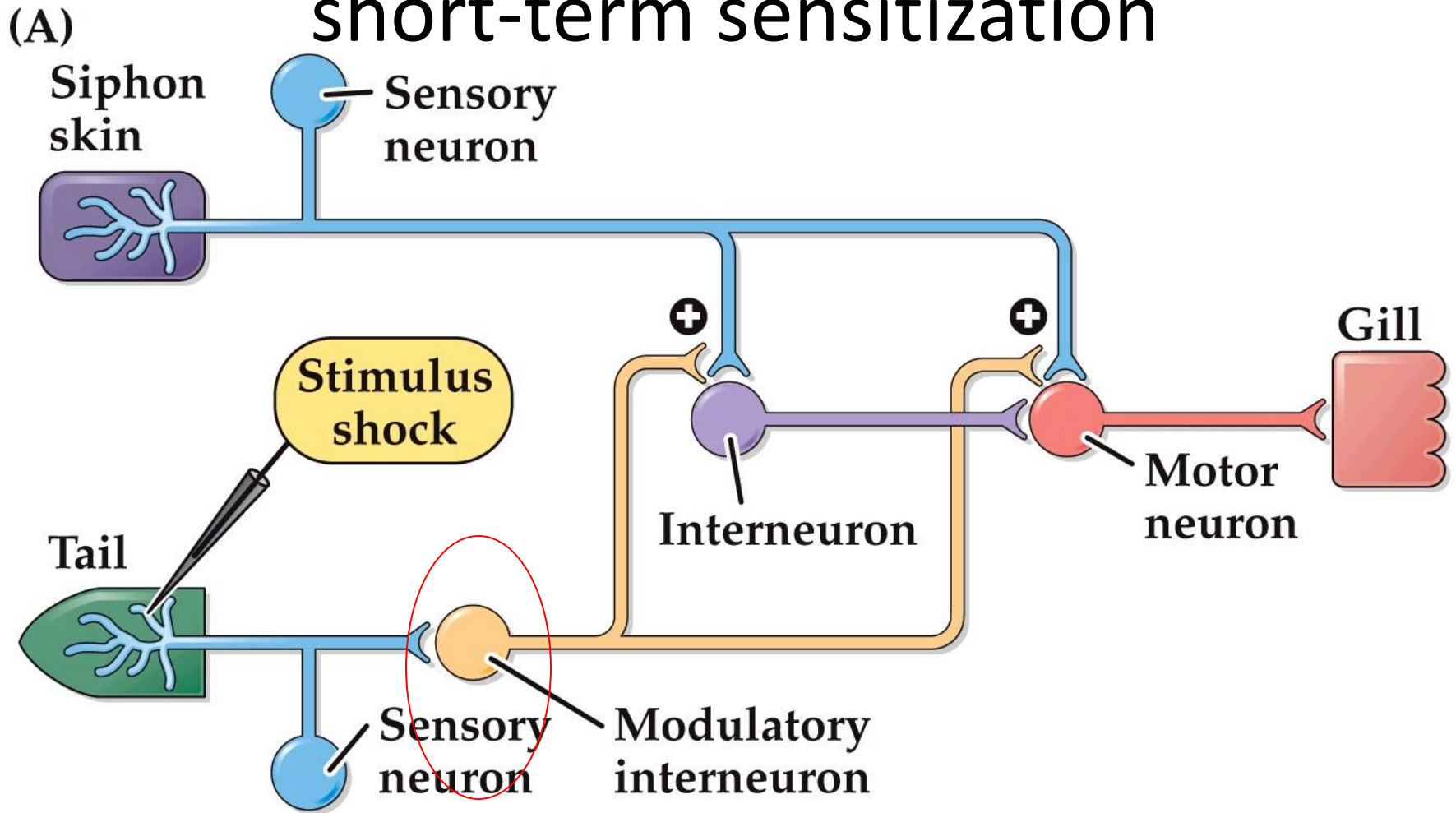
Sensitization or habituation?



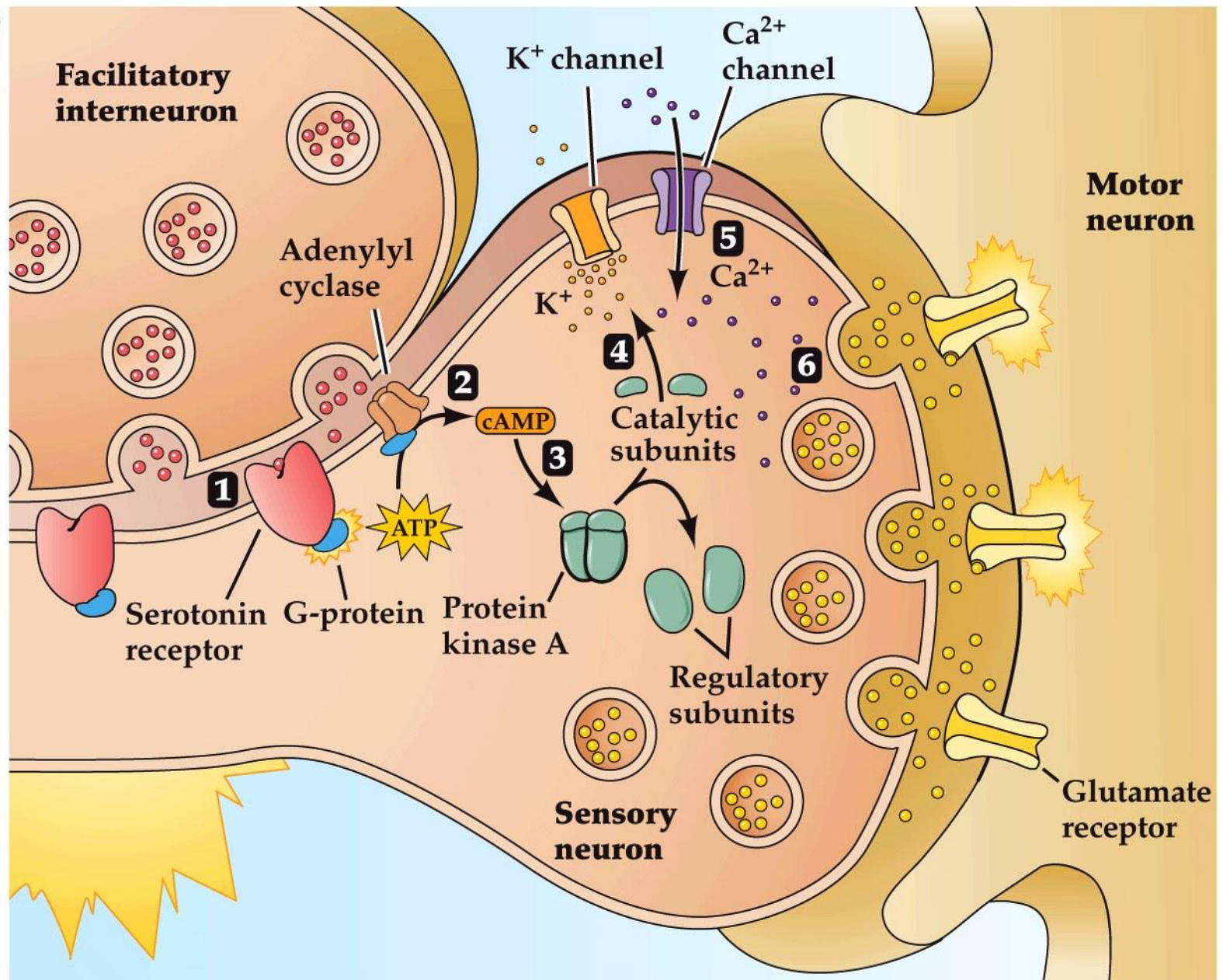
Sensitization



Synaptic mechanisms underlying short-term sensitization



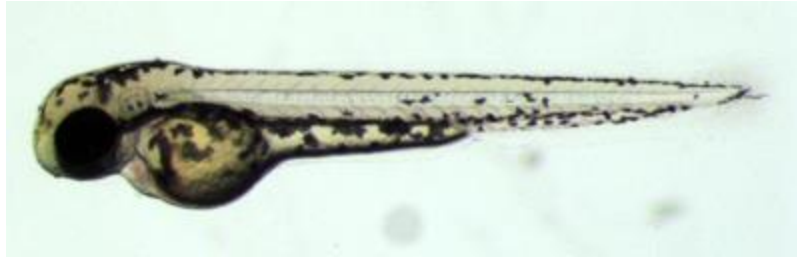
(A)



Eric Kandel

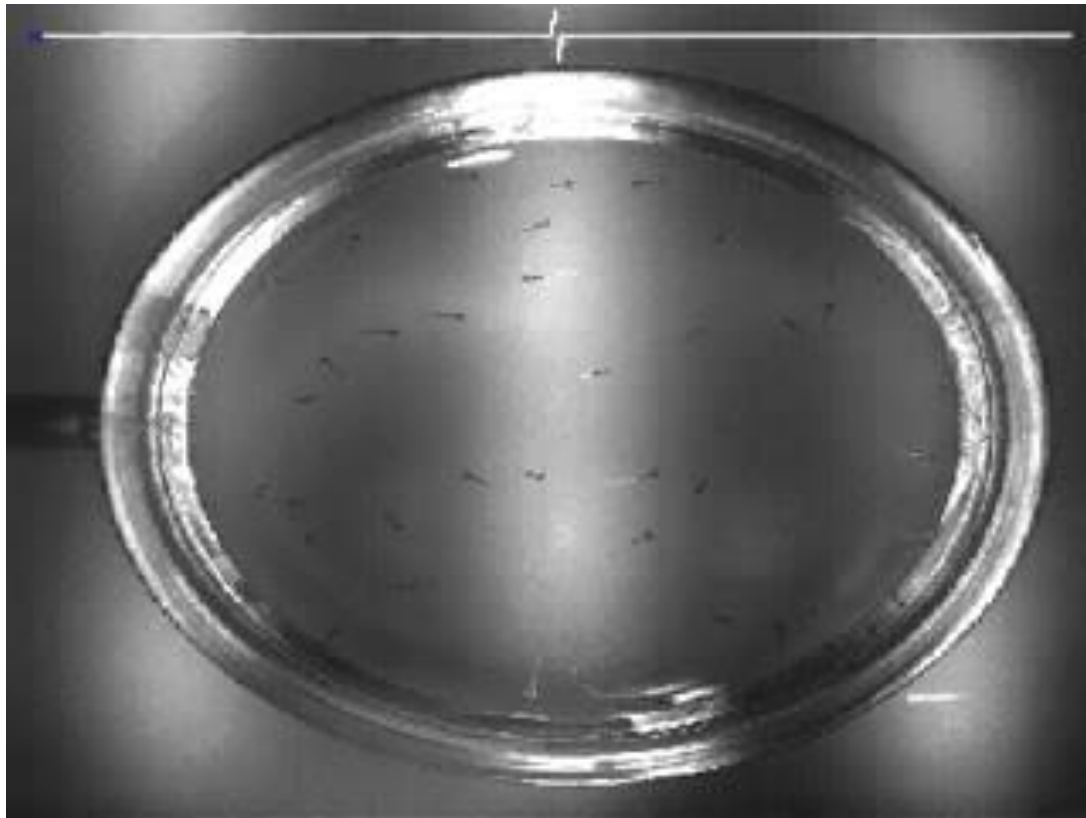
- Nobel Prize in Physiology or Medicine
- Prize motivation: "for their discoveries concerning signal transduction in the nervous system"
 - Discovery of phosphorylation in nervous system.
With DARPP-32 protein
 - Paul Greengard & Arvid Carlsson shared prize

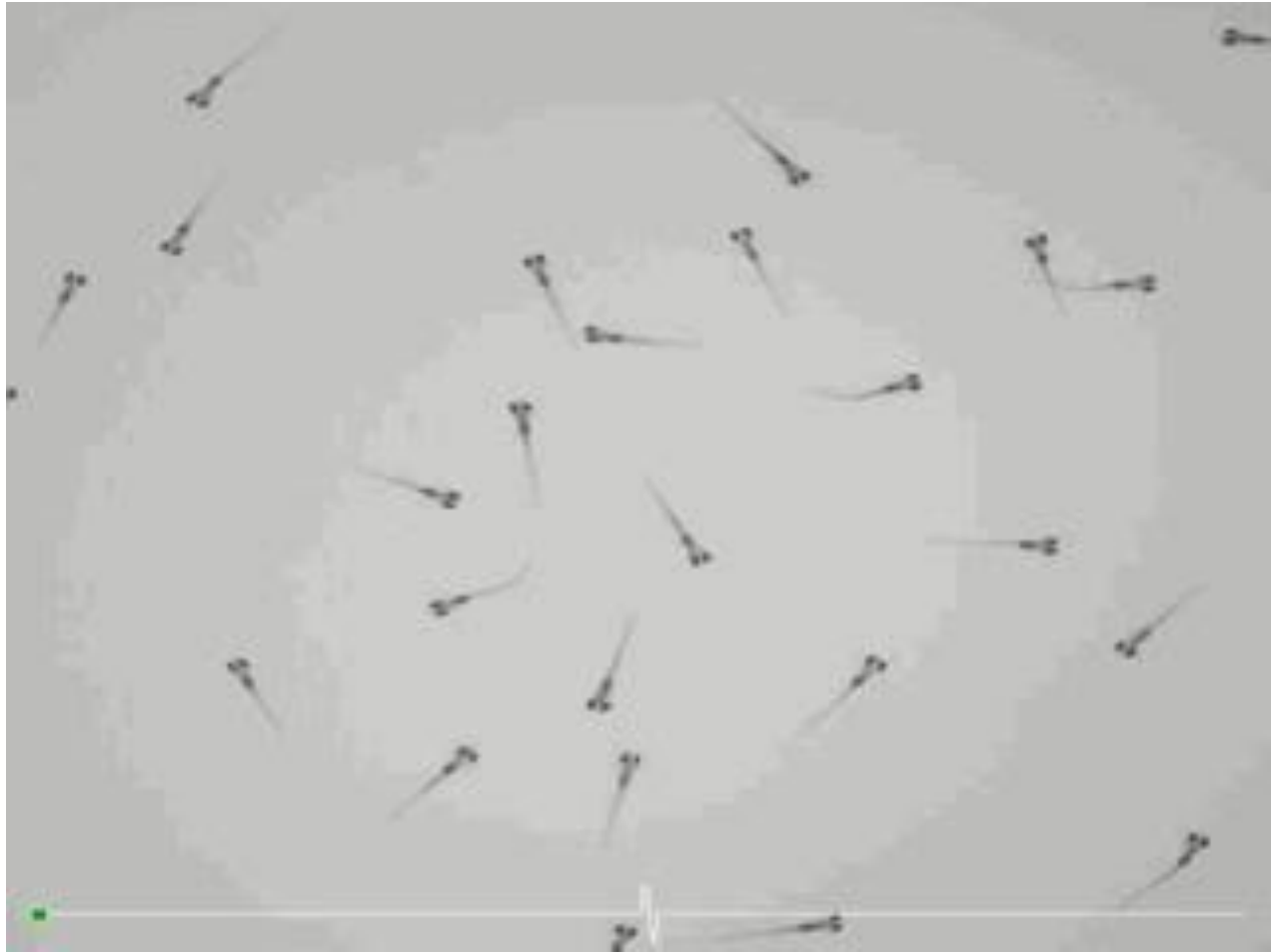
Zebrafish and learning



Startle Response

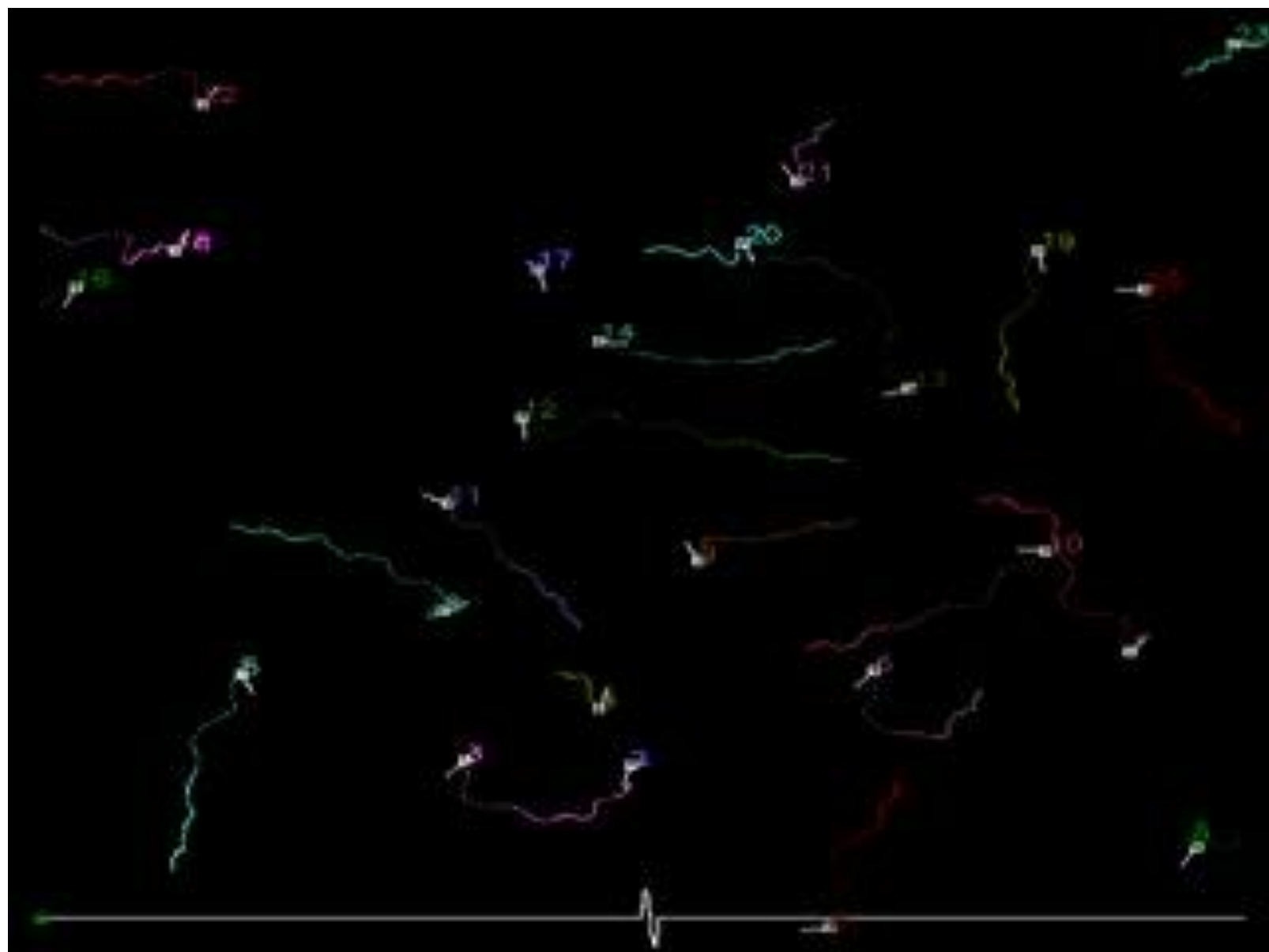
- A defensive reaction to a sudden, unexpected stimulus

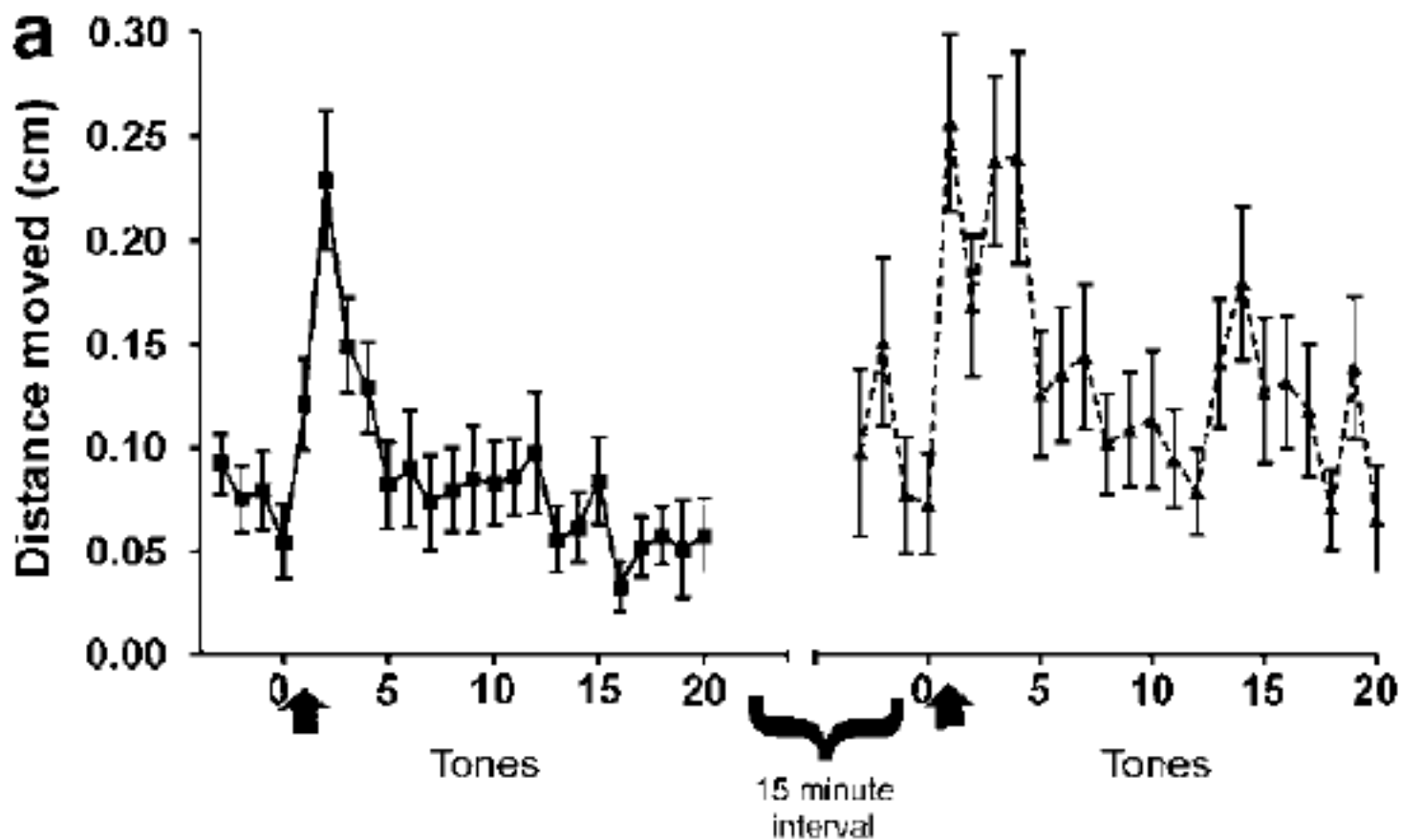




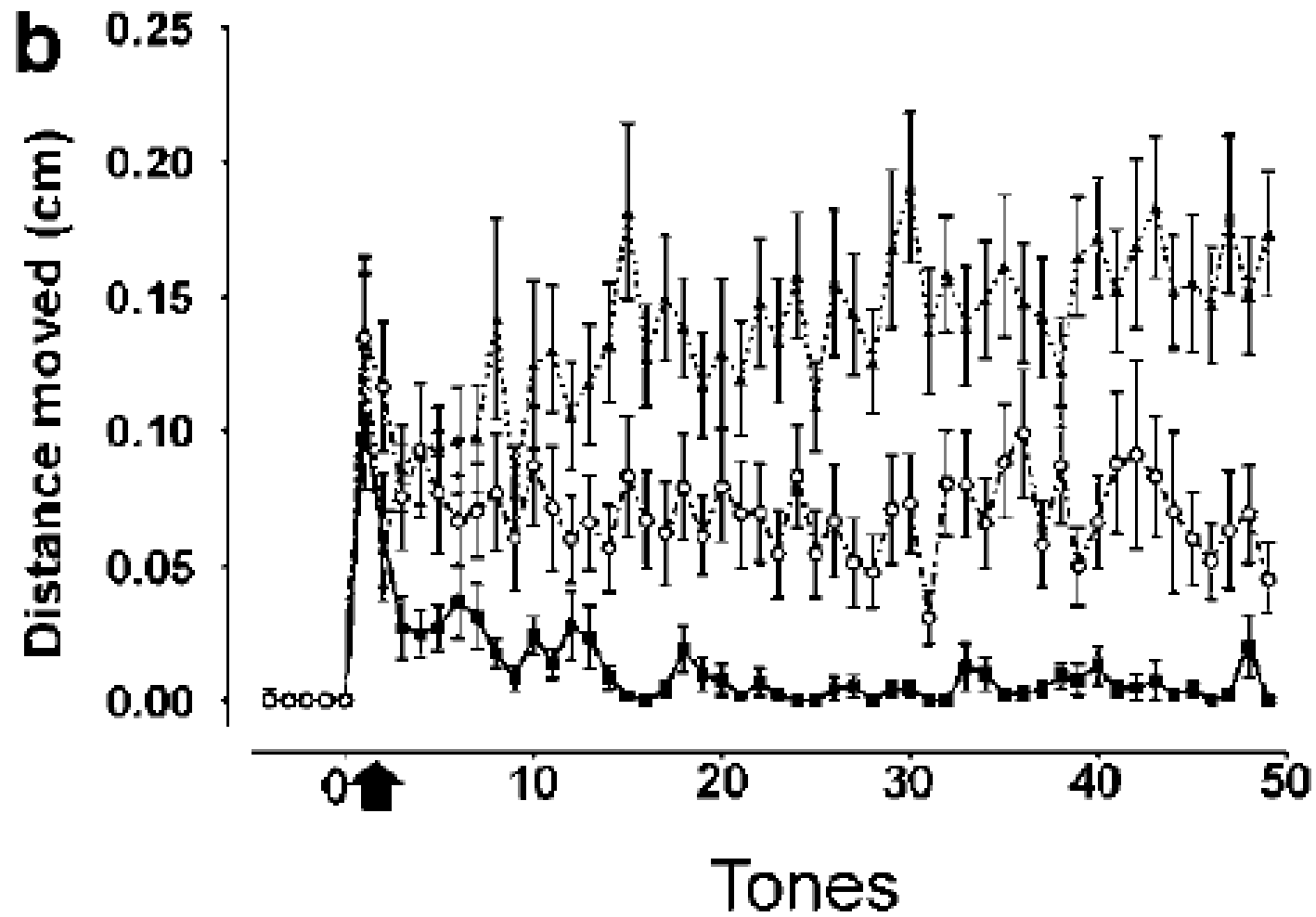
How would you quantify this
behavior?



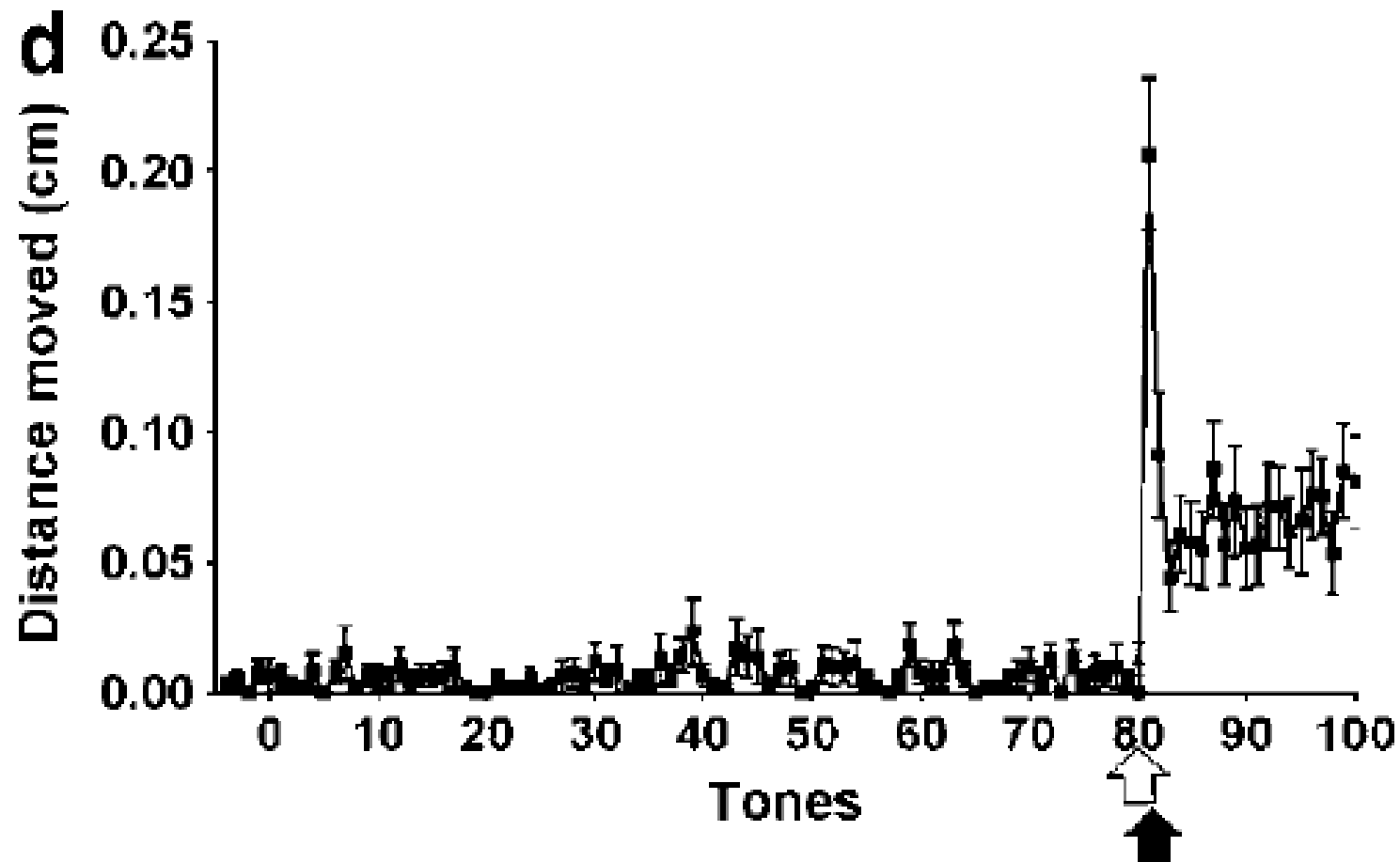




1,5,20S ITI



Dishabituation



Article

Dopamine-based mechanism for transient forgetting

<https://doi.org/10.1038/s41586-020-03154-y>

Received: 3 April 2020

Accepted: 9 December 2020

Published online: 20 January 2021



Check for updates

John Martin Sabandal¹, Jacob A. Berry¹ & Ronald L. Davis¹✉

Active forgetting is an essential component of the memory management system of the brain¹. Forgetting can be permanent, in which prior memory is lost completely, or transient, in which memory exists in a temporary state of impaired retrieval. Temporary blocks on memory seem to be universal, and can disrupt an individual's plans, social interactions and ability to make rapid, flexible and appropriate choices.

- How could you use the principles of habituation and sensitization to address problems like food allergies or chronic pain?

FIRST OPINION

Oral immunotherapy can be a game changer for many with food allergies

By CHARLES FENG / DECEMBER 11, 2019



PATRICK SISON/AP

Food allergies upend lives. In children, food allergies can scale back or completely stop participation in birthday parties, organized sports, and

STAT+

Exclusive biopharma, health policy, and life science analysis. Start your 30-day free trial today.

TRY STAT PLUS

MOST POPULAR



New coronavirus can cause infections with no symptoms and sicken otherwise healthy people, studies show



The basics: What we know — and don't know — about the virus spreading in China and beyond



Four 'generations' of spread seen with virus in China, alarming experts



Former CDC director Tom Frieden on 3 key questions



TREATMENTS

A New Treatment Can Relieve Food Allergies, But Few Doctors Offer It

February 21, 2019 · 2:54 PM ET

ESTHER LANDHUIS



Some patients have started to seek out an oral treatment for food allergies that's not yet widely offered by clinicians.

Cal Gwynn/Getty Images

Opponent-Process Theory of Emotions

- Explains the aftereffects of strong emotions.

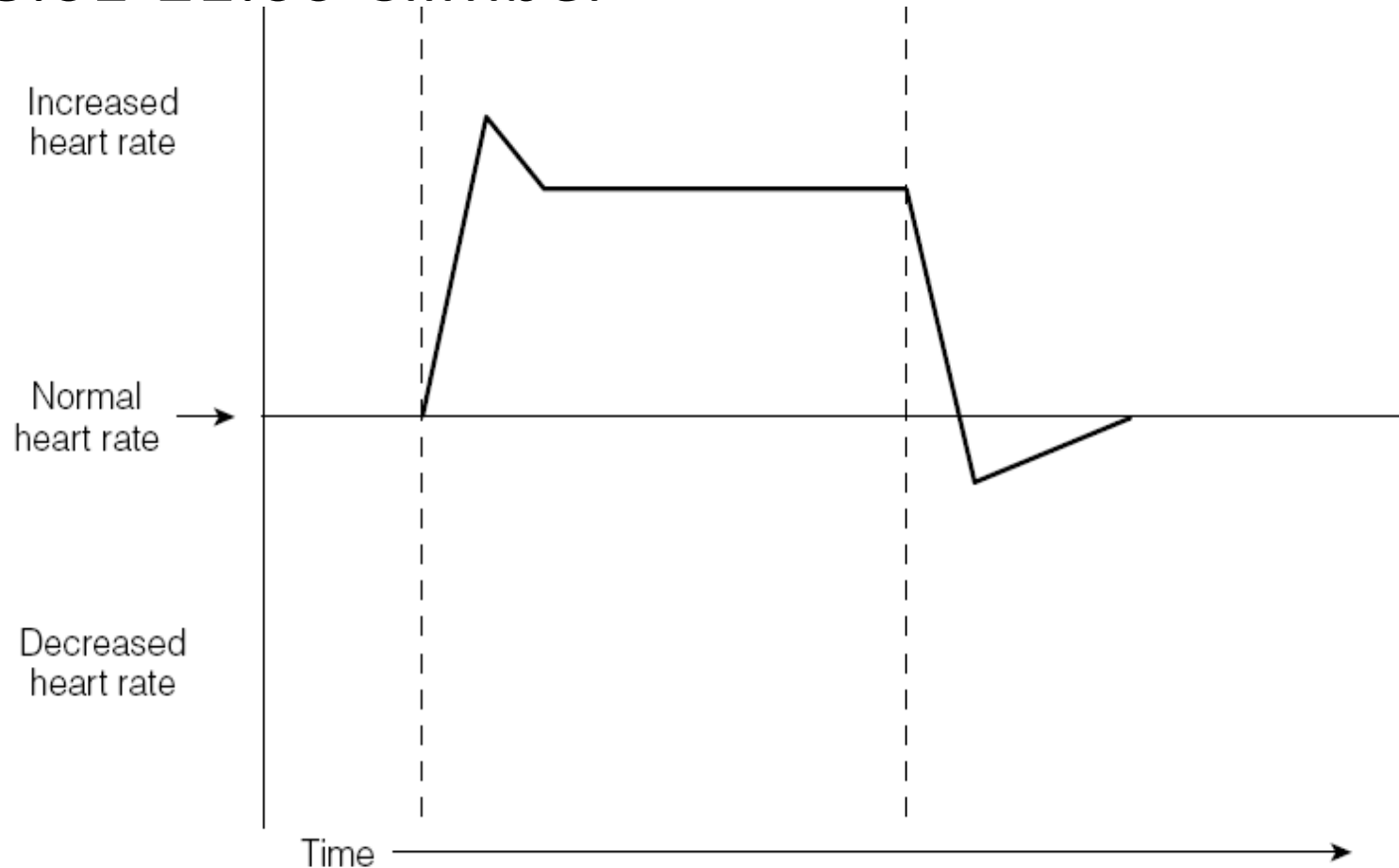


Opponent-Process Theory of Emotions

- An emotional event elicits two competing processes:
 - an a-process (or primary process) that is directly elicited by the event, and
 - a b-process (or opponent process) that is elicited by the a-process and serves to counteract the a-process.

The Moth: Live from Portsmouth

- 18:01-21:00 Climber



The A-Process

- Correlates closely with the presence of the emotional event
- When the height is presented, heart rate immediately increases; when the height is removed, heart rate immediately decreases.

The B-Process

- is slow to increase and slow to decrease.
- Slowly the b-process begins to moderate the a-process, causing a slight decrease in heart rate before stabilizing.
- When the height is removed, the a-process immediately disappears; but the b-process only slowly declines.

What happens when the climber repeatedly climbs?

- With repeated presentations of the emotional event, the b-process increases in both strength and duration.
- The increase in heart rate during each exposure to high heights becomes less and less extreme.
- Each time the removal of the height occurs, the climber's heart rate plunges more and more deeply and takes increasingly longer to return to normal.

Additional Examples

Military parachutists became less and less terrified with repeated jumps and became more and more elated following each jump.

Drug Abuse

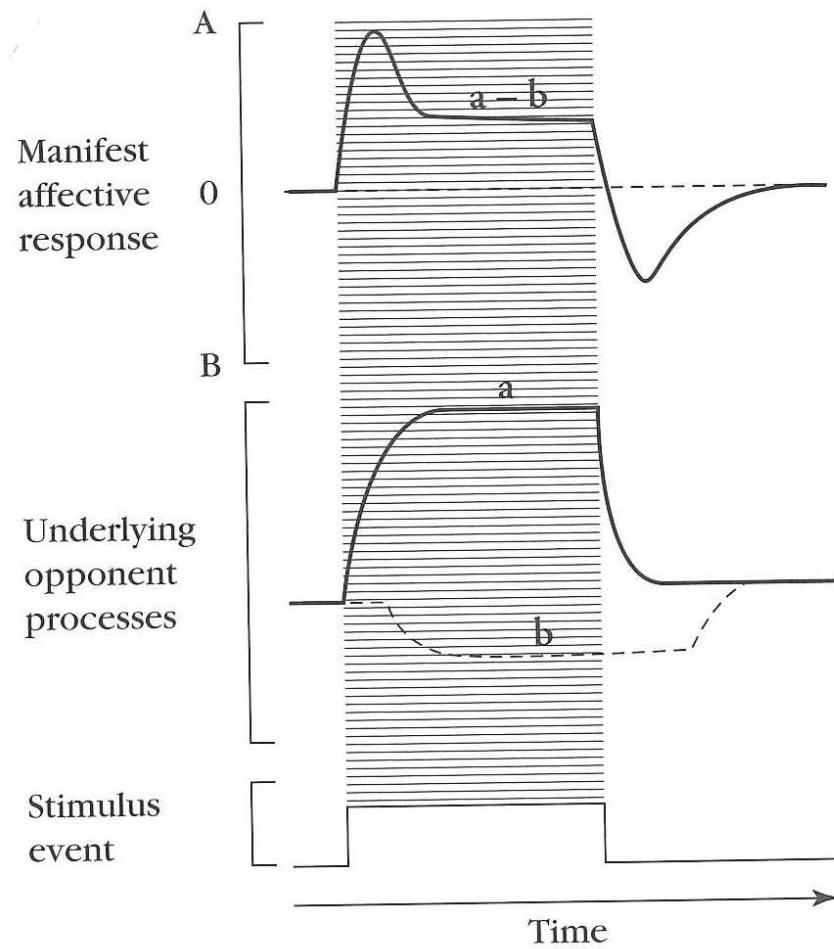


FIGURE 2.15

Source: Adapted from Solomon, R. M. (1980). The opponent process theory of emotion.

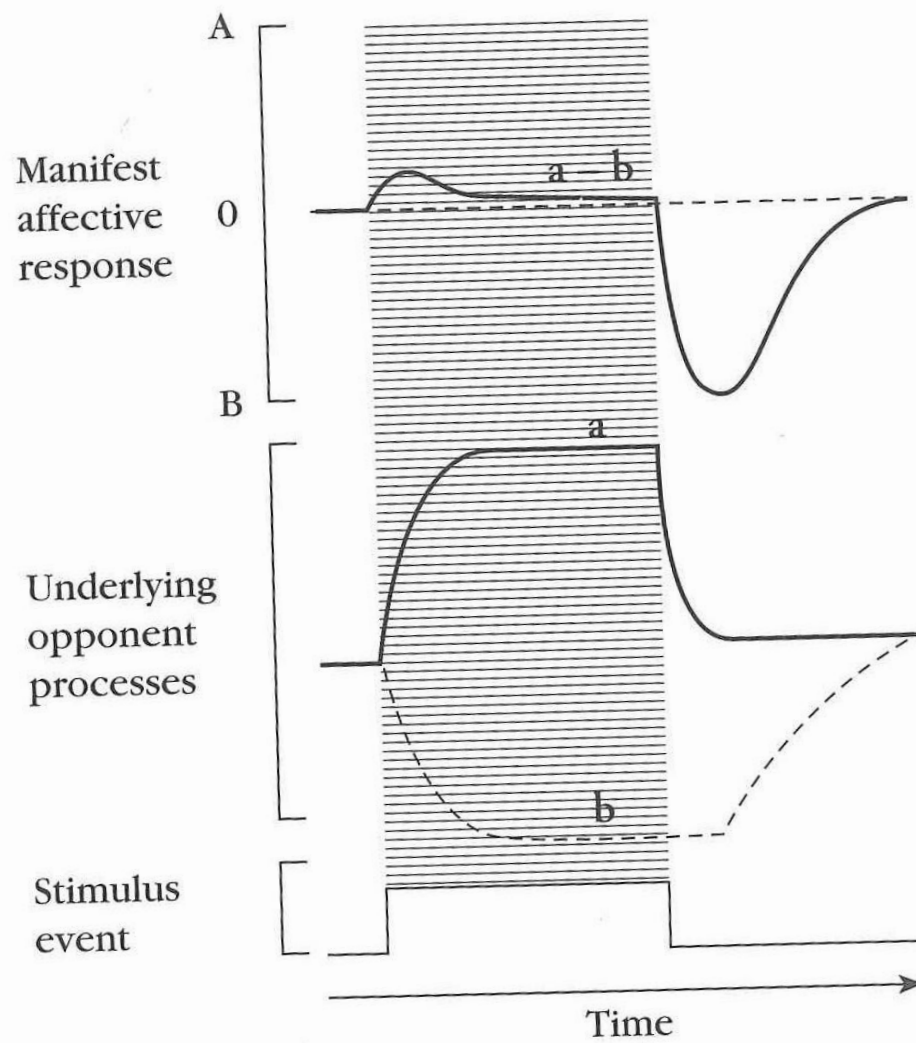


FIGURE 2.16

What are some examples of opponent-process theory of emotion?

- Public speaking
- Scary movies
- Boredom after sporting event



©2012 Joel A. Rogers
www.CoasterGallery.com



- Are events that are initially terrifying/fearful when repeated could last in long-term euphoria?
- How can you prevent against emotional sensitization?

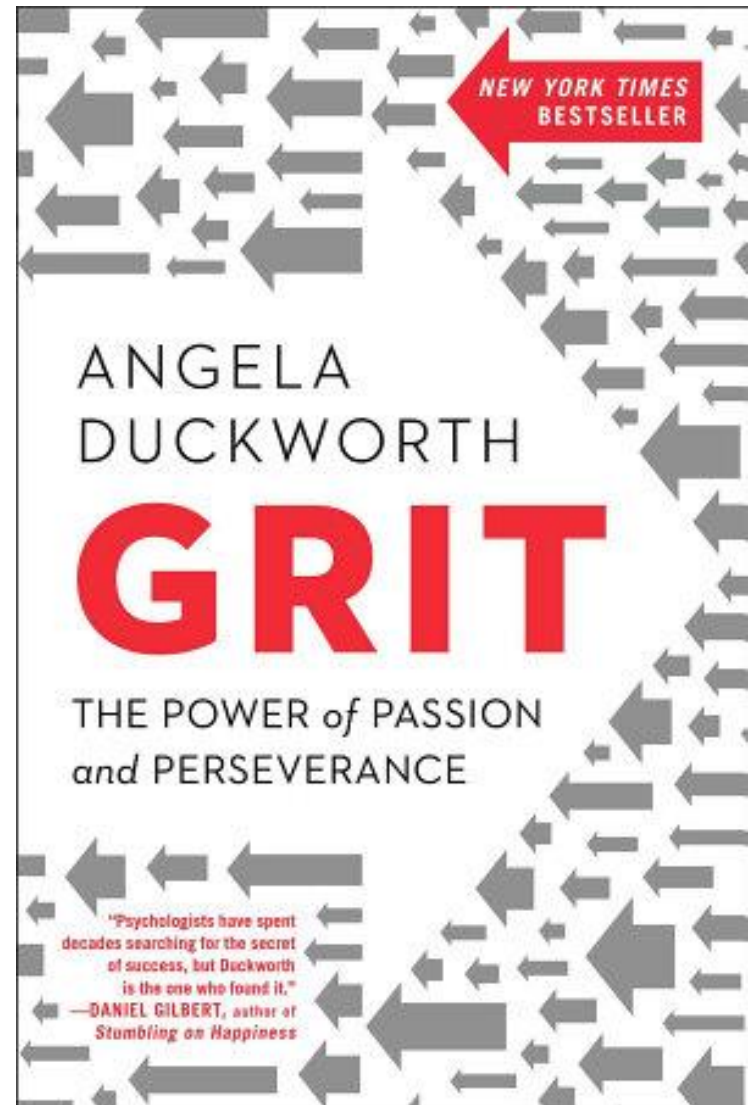
#1 New York Times bestseller

The Road to Character



DAVID
BROOKS

Author of *The Social Animal*



Learning – Our survival depends on it!

- Our environment changes constantly and is filled with a variety of stimuli.
- We need a mechanism by which we can adjust our behavior to changing conditions.
- Biological or evolutionary change is too slow.

Learning –

Our survival depends on it! **Continued...**

- The ability to relate events to each other allows us to better anticipate the future, thereby greatly facilitating our chances of surviving.

Chemical modulation of memory formation in larval zebrafish

Marc A. Wolman, Roshan A. Jain, Laura Liss, and Michael Granato¹

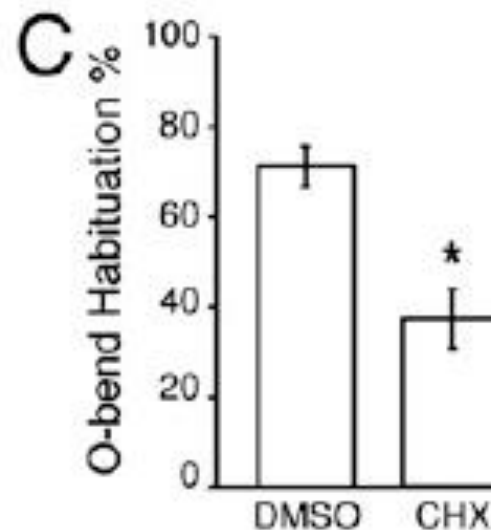
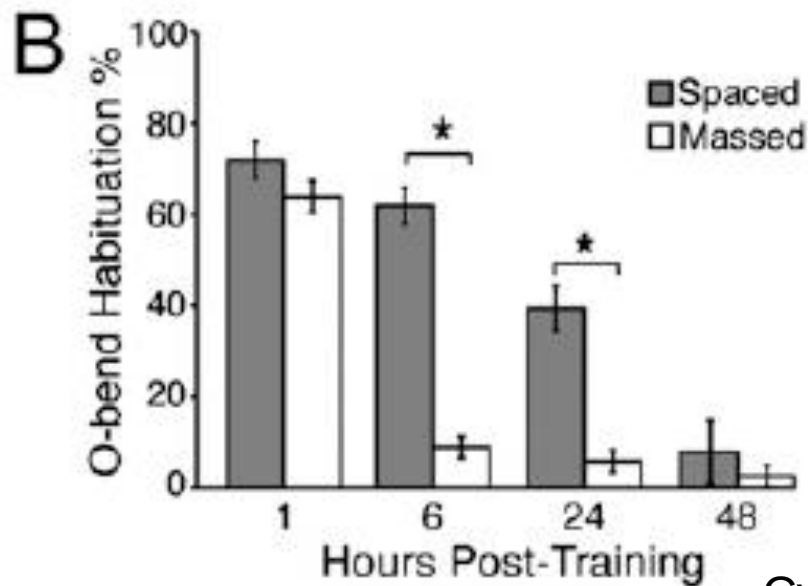
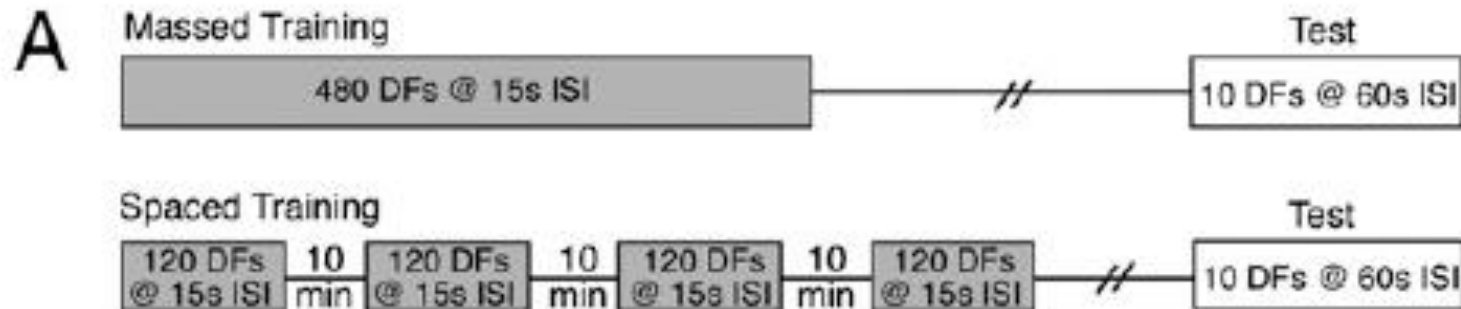
Department of Cell and Developmental Biology, University of Pennsylvania School of Medicine, Philadelphia, PA 19104

Edited* by John E. Dowling, Harvard University, Cambridge, MA, and approved August 5, 2011 (received for review May 12, 2011)

Whole organism-based small-molecule screens have proven powerful in identifying novel therapeutic chemicals, yet this approach has not been exploited to identify new cognitive enhancers. Here we present an automated high-throughput system for measuring nonassociative learning behaviors in larval zebrafish. Using this system, we report that spaced training blocks of repetitive visual

as well as the time required to train and test large numbers of animals (10–16). Larval zebrafish execute a repertoire of simple, well-defined, and stereotyped sensorimotor behaviors that have accessible and characterized circuitry and provide a vertebrate system amenable to large-scale forward genetic and chemical screening (17–24). Although some studies have explored the

Massed vs. Spaced Training



Cyclohexamide-blocks protein synthesis

- The authors move away from the O-bend and towards a different measure. What measure for habituation did they use?

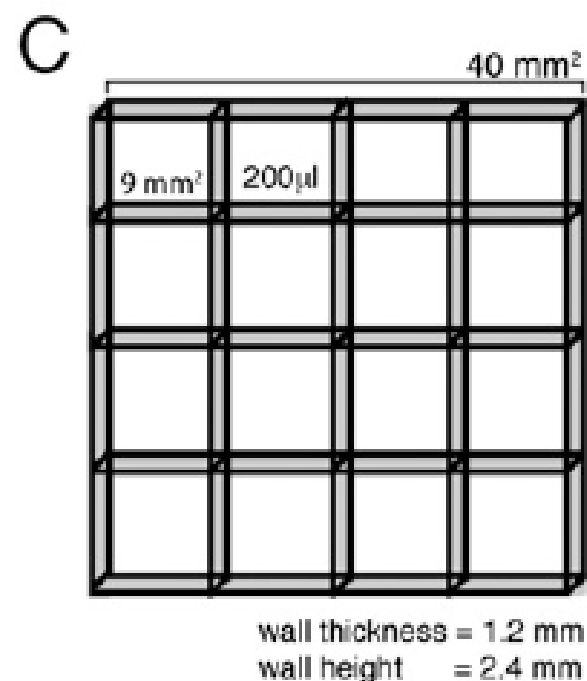
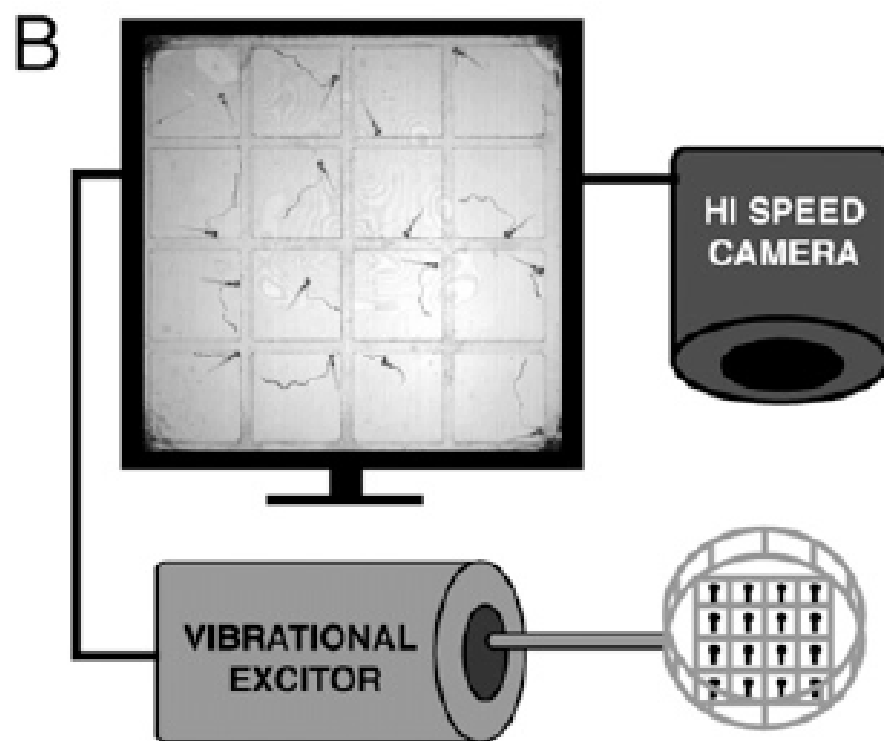
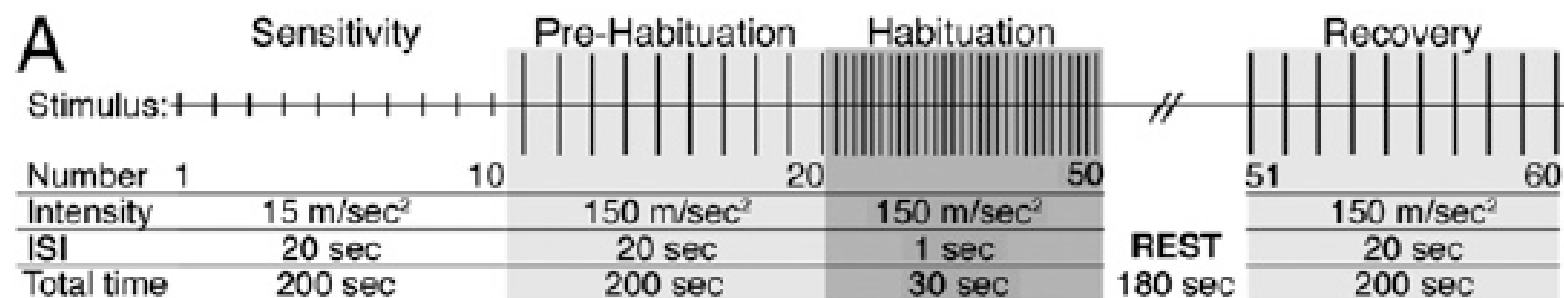
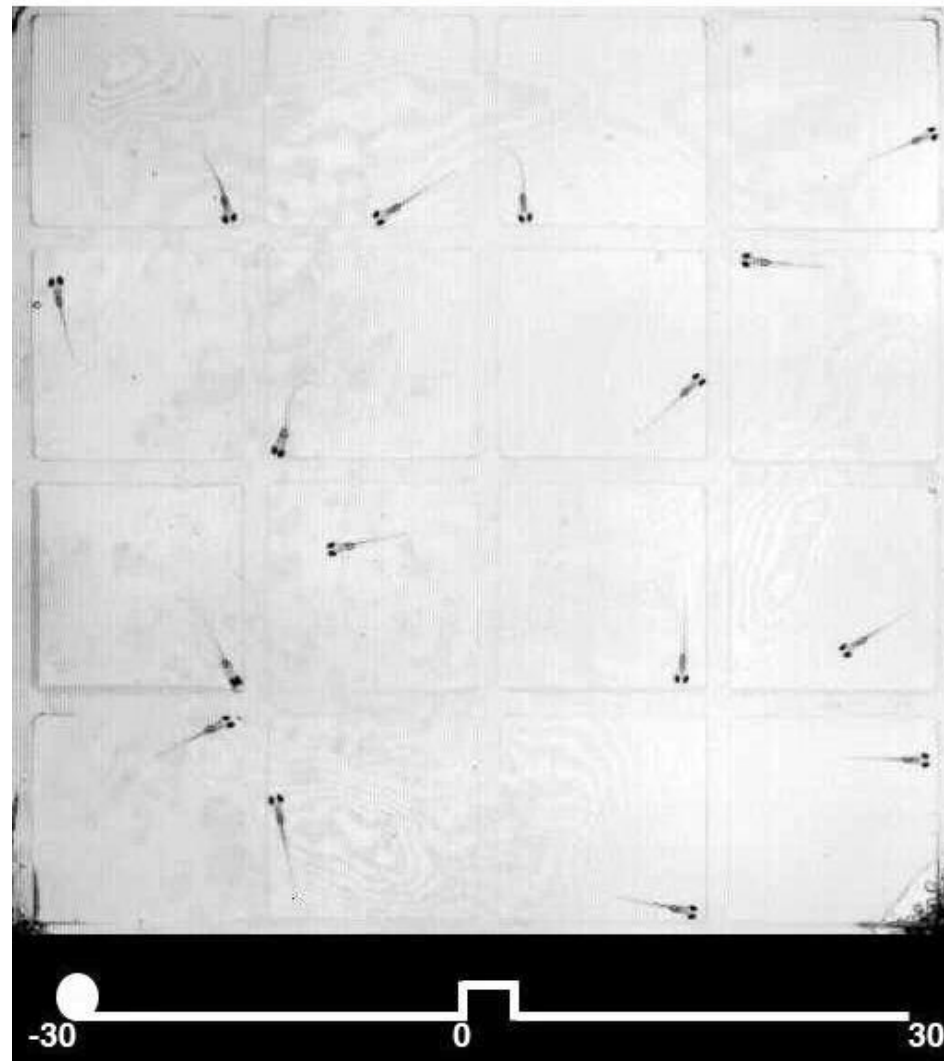
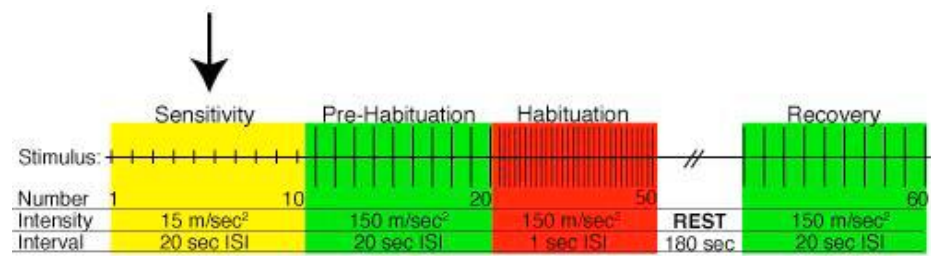
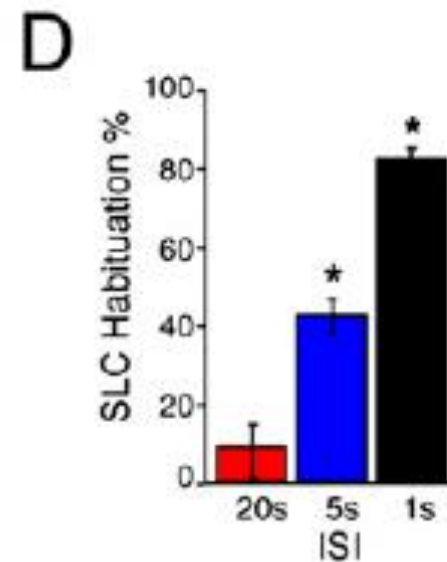
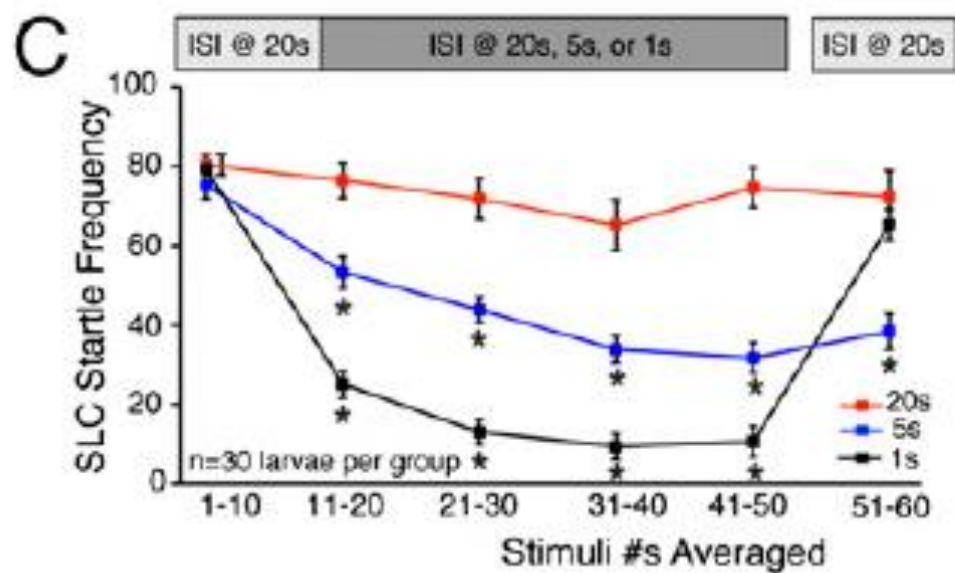
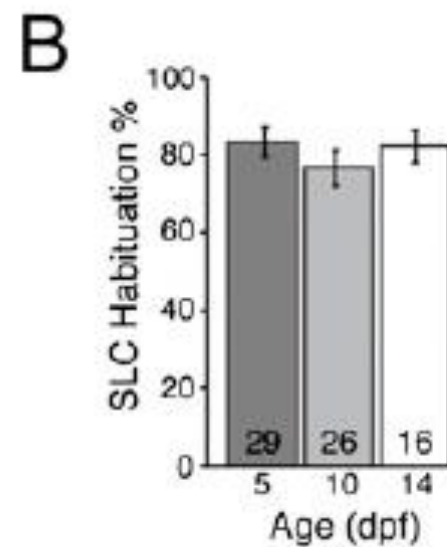
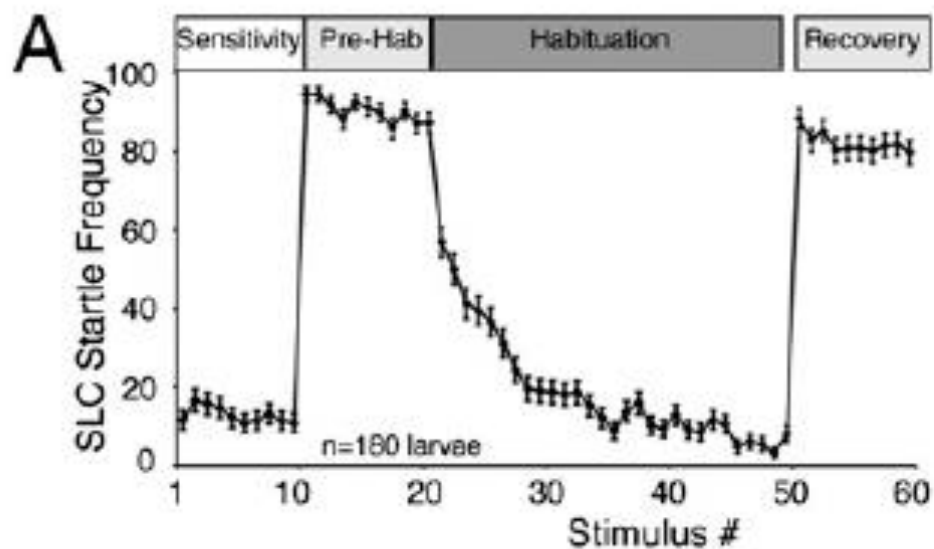
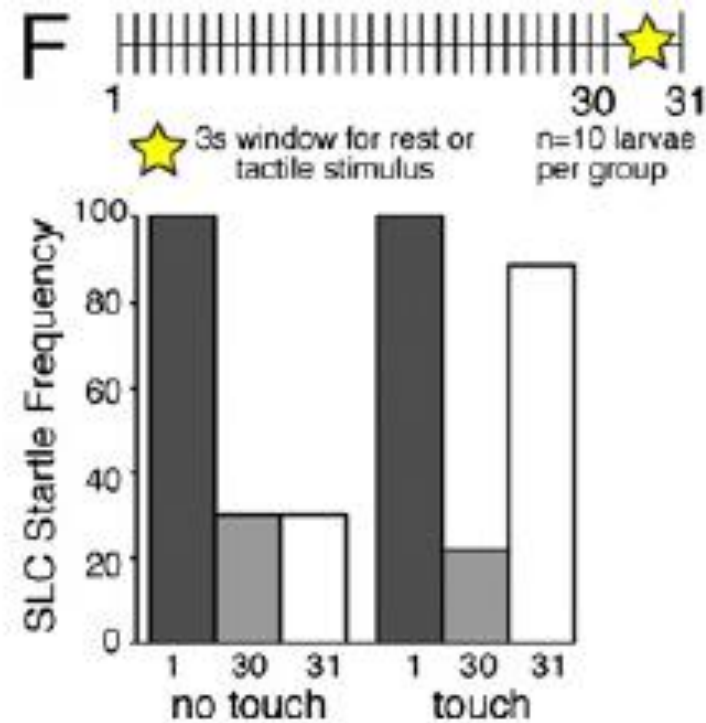
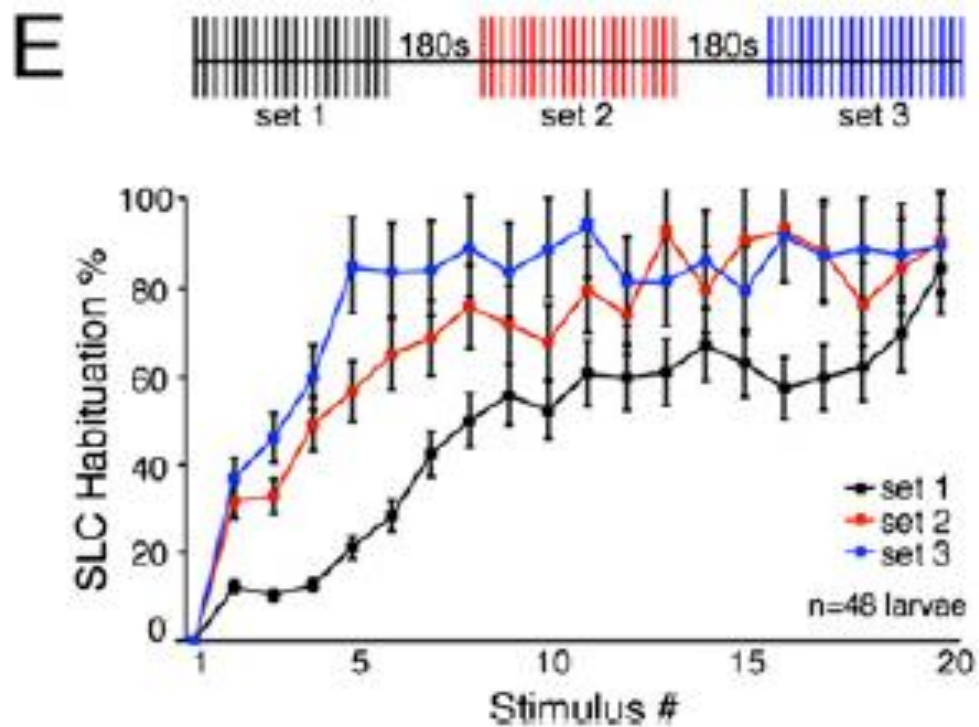


Fig. 2. Acoustic startle habituation assay and testing apparatus. (A) Larvae

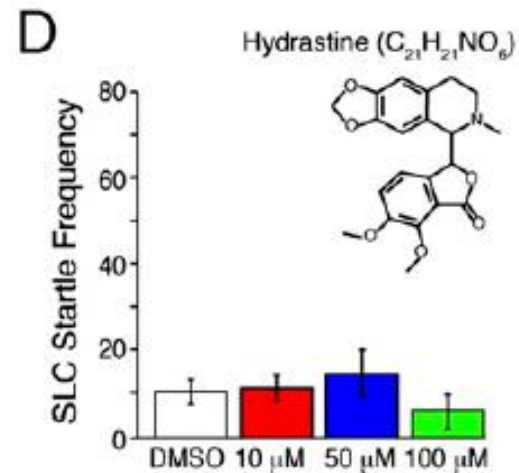
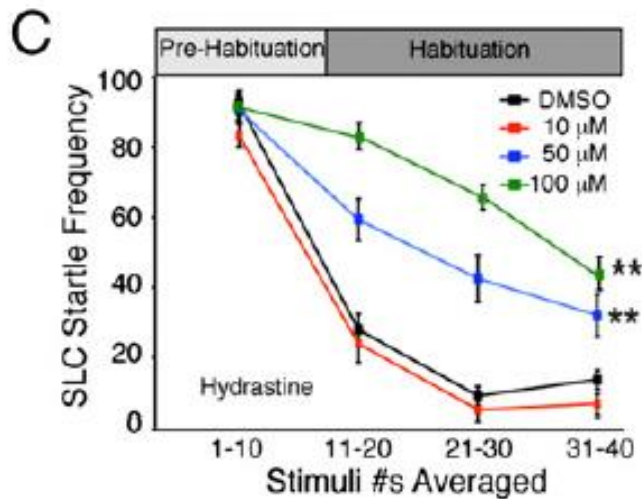
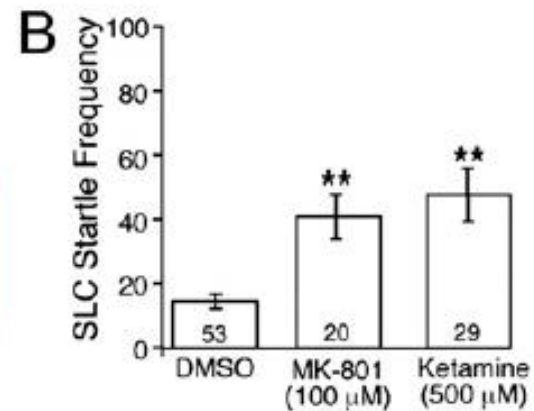
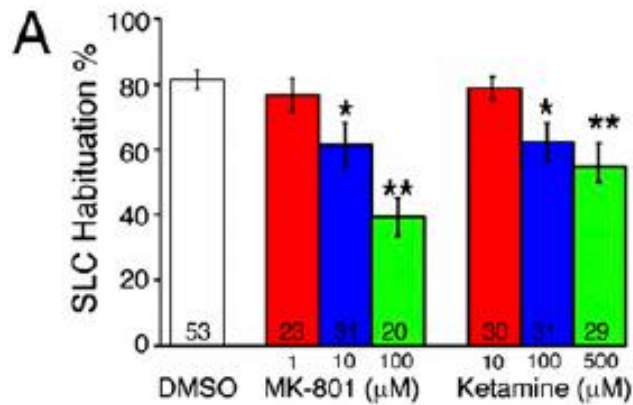






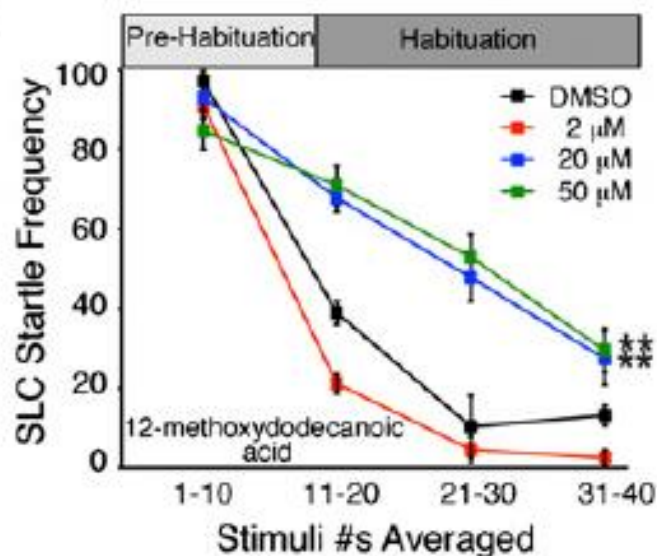
Describe the process here

Drug Intervention

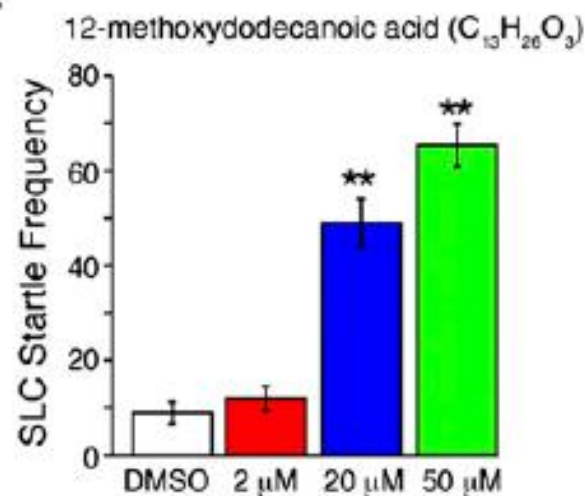


- MK-801, is a noncompetitive antagonist of the N-Methyl-D-aspartate receptor, a glutamate receptor

E

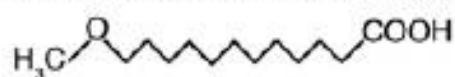


F

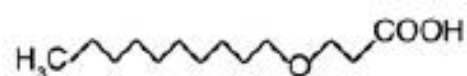


G

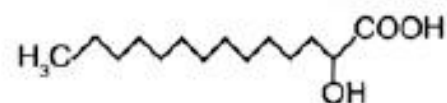
12-methoxydodecanoic acid ($C_{13}H_{26}O_3$)



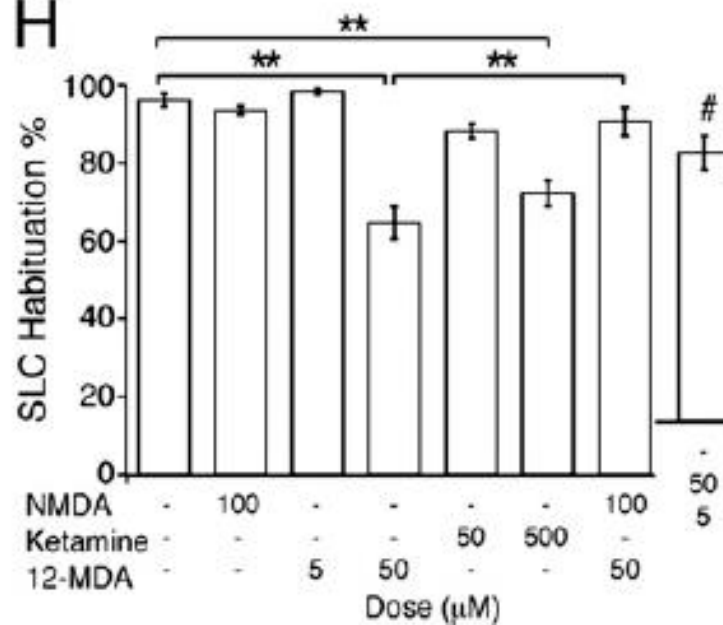
4-oxatetradecanoic acid ($C_{13}H_{26}O_3$)



2-hydroxymyristic acid ($C_{14}H_{28}O_3$)

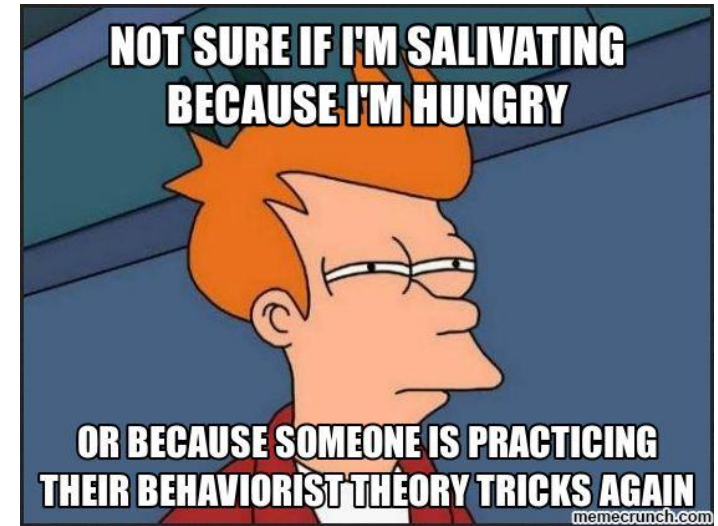
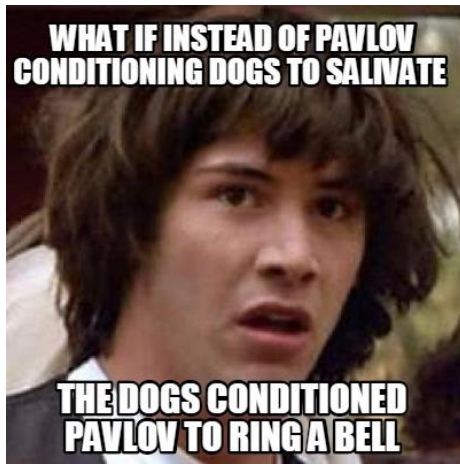


H



Summary Points

- Dark flashes result in habituation of O bend
- A 150 mm²/sec acoustic stimulus with an intertrial interval of 1 s will result in significant habituation
- A hand held poker will result in dishabituation
- NMDA antagonists will reduce habituation



Classical Conditioning



Classical Conditioning

- One mechanism for learning
- Also called Pavlovian or respondent conditioning
- One stimulus that does not elicit a response is associated with a second stimulus that does.
- As a result, the first stimulus also comes to elicit a response.

Pavlov's Discovery

- Ivan P. Pavlov (1849–1936)
- a Russian physiologist
- generally credited with the first systematic investigations into classical conditioning
- investigated salivation, the initial step in the digestive process
- found that salivation could be initiated with “psychic” factors

Pavlov's Basic Procedures of Classical Conditioning

- Purpose: to train a dog to salivate to the sound of a metronome.
- The dog was restrained in a harness, and a tube was inserted into an incision that had been made in its cheek.
- The saliva would run down the tube into a container where it could be precisely measured.

Unconditioned Stimulus (US)

- Defined as “a stimulus that naturally elicits a response.”
- The food is called an *unconditioned stimulus* (US).
- The dog did not have to learn to respond to the food.



Unconditioned Response (UR)

- Defined as “the response that is naturally elicited by the US”
- Before conditioning, the dogs would automatically salivate in response to the taste of food.



Neutral Stimulus

- The sound of a metronome, however, does not elicit salivation and is therefore said to be a *neutral stimulus* (NS) with respect to salivation.
- The neutral stimulus does not elicit a response.



During Conditioning

- The sound of the metronome is presented just before the food.
- The dog continues to salivate in response to the food.
- The sound of the metronome and the food are presented several times.

After Conditioning

- The sound of the metronome elicits salivation.
- The food does not have to be present for the dog to salivate when it hears the sound of the metronome.

Conditioned Stimulus

- Defined as “any stimulus that, although initially neutral, comes to elicit a response because it has been associated with an unconditioned stimulus”
- The dog is now salivating to the sound of the metronome and this response required training (conditioning).
- The sound of the metronome is now called a *conditioned stimulus* (CS).

Conditioned Response

- Defined as “the response, often similar to (but never identical to) the UR, that is elicited by the CS”
- Salivating in response to the sound of the metronome is now called a *conditioned response* (CR).

Schematic Diagram: Before Conditioning

Food	→	<i>Salivation</i>
US		UR

Metronome	→	No salivation
NS		—

Schematic Diagram: During Conditioning

Metronome:	Food	→ <i>Salivation</i>
NS (or CS)	US	UR

- Each pairing of the NS and US during conditioning is called a *conditioning trial*.

Schematic Diagram: After Conditioning

Metronome → *Salivation*

CS

CR

Another Example

- A child is bit by a dog.

Dog: Bite → *Fear*
NS US UR

Dog → *Fear*
CS CR

What are Other Examples?

NS : US → UR

CS → CR

Describing Classical Conditioning

- Often classical conditioning procedures fall into two categories: appetitive and aversive.
- We also talk about classical conditioning being excitatory or inhibitory.

Appetitive Conditioning

- The US is an event that is usually considered pleasant and that an organism seeks out.
- Examples:
 - *food, water, sexual stimuli*



Aversive Conditioning

- The US is an event that is usually considered unpleasant and that an organism usually avoids.
- Examples:
 - *an electric shock, a painful bite, and an unpleasant odor*
- Usually can occur quickly and requires only a few pairings to be conditioned
- To survive, we have evolved in such a way as to quickly learn to dislike those events that cause pain or illness.

Measuring Fear

- *Conditioned suppression.*
- The level of fear can be assessed by the extent to which behaviors stop when fear-invoking stimuli is present and resumes when it is not.
- This is measured in the form of a *suppression ratio*.



Suppression Ratio

$$\text{Suppression Ratio} = \frac{\text{\# of CS responses}}{\text{\# of CS responses} + \text{\# of pre-CS responses}}$$

Example

- The rat is first trained to press a lever to obtain food.
- When a steady rate of lever pressing has been established, a fear conditioning procedure is introduced.
- A 30-second tone is presented followed by a 1-second shock. Thus:

30" Tone:	1" Shock	→	<i>Fear</i>
NS	US		UR

After Conditioning

- As conditioning proceeds, however, the tone too will come to elicit fear, and the rat will stop pressing the lever when it hears the tone.

30" Tone → *Fear*

CS

CR

Measuring the Rat's Fear

- The rat emits 20 responses during the 30-second pre-CS period followed by 0 responses during a 30-second CS period.0

$$0/(0 + 20) = 0$$

- A suppression ratio of 0 indicates total suppression of responding.

What if the rat...?

- What if the rat emitted 10 responses during the CS period?

$$10 / (10 + 20) = 10 / 30 = .33$$



What if the rat...?

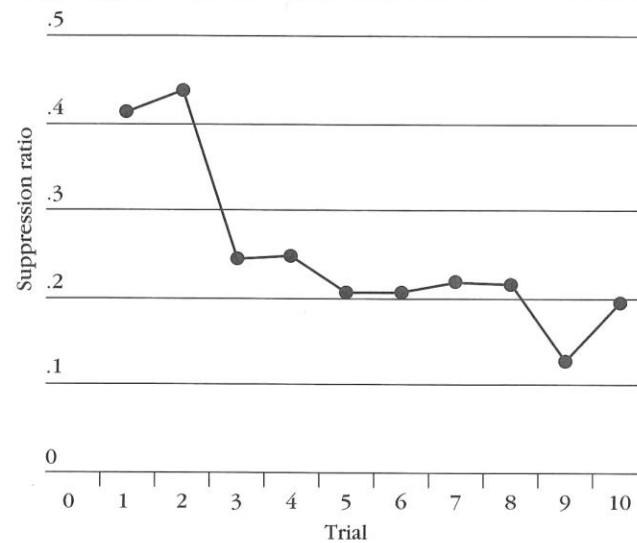
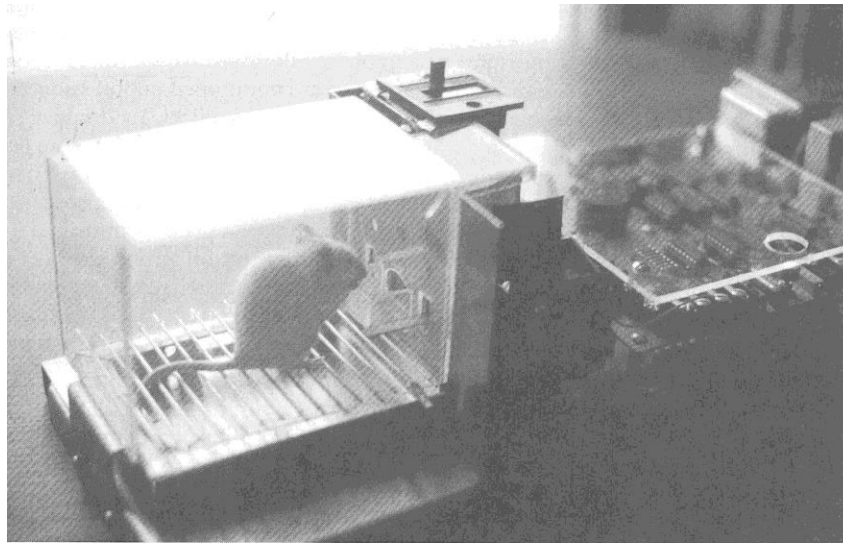
- What if the rat's behavior didn't change?

$$20 / (20 + 20) = 20 / 40 = .50$$

- The suppression ratio will generally vary between 0 and .5, with a lower ratio *indicating less responding*, and *less responding indicating greater suppression and more effective conditioning*.



Conditioned Suppression



B

Excitatory and Inhibitory conditioning

- Excitatory conditioning
 - The NS is associated with the presence of the US
 - Bell and salivation
- Inhibitory conditioning
 - The NS is associated with the absence of the US
 - Presence of owner is inhibitory for fear
 - Light and no food

Temporal Arrangement of Stimuli

- There are several ways to temporally arrange the presentation of the NS and the US:
 - Delayed Conditioning.
 - Trace Conditioning.
 - Simultaneous Conditioning.
 - Backward Conditioning.

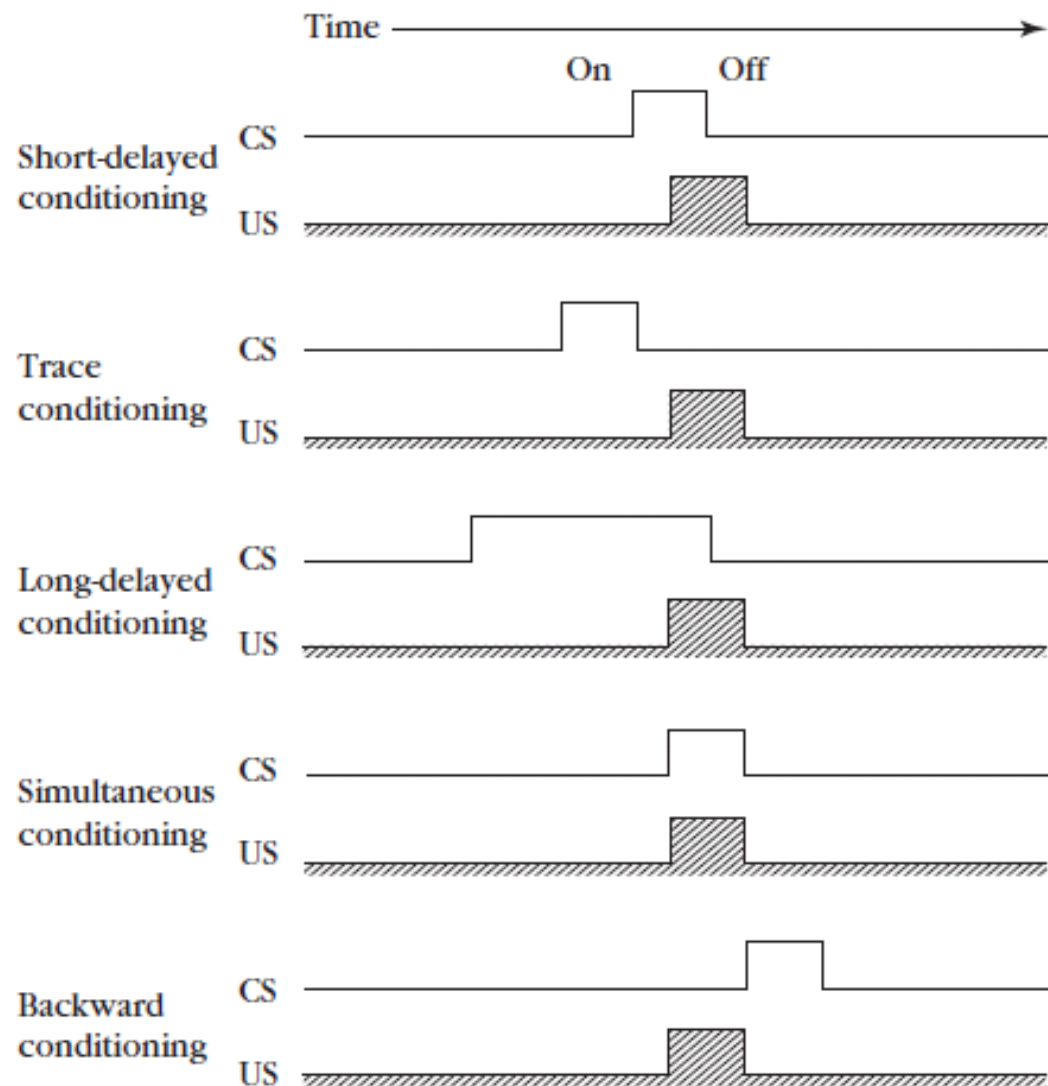
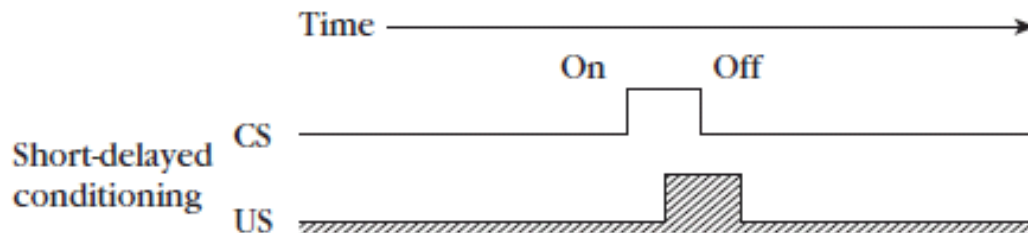


FIGURE 3.7

Five common classical conditioning procedures.

Delayed Conditioning

- The onset of the NS precedes the onset of the US, and the two stimuli overlap.
- Example:
 - *first present the tone and then, while the tone is still on, present a shock*
- best arrangement for conditioning, especially when the *interstimulus interval (ISI)* is relatively short.
- *The interstimulus interval (ISI)* is the time between the onset of the NS and the onset of the US.



Trace Conditioning

- The onset and offset of the NS precede the onset of the US.
- The NS occurs before the US, and the two stimuli do not overlap.
- the organism has to “remember” the occurrence of the tone (“memory trace”) to be able to associate the NS and the US.
- Example:
 - *a tone is turned on and then off, and this is then followed by the presentation of a shock*

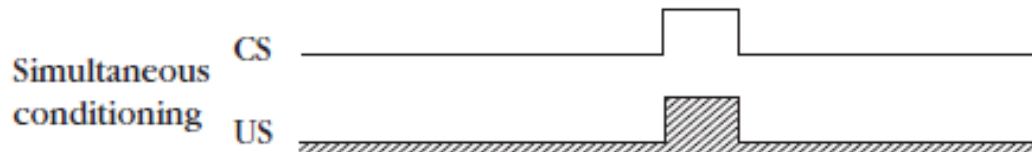


Trace Vs. Delayed Conditioning

- Trace conditioning can be almost as effective as delayed conditioning.
- The trace interval has to be relatively short (no more than a few seconds).
- If the trace interval is longer than that, conditioning is unlikely to occur.
- *Trace interval* is the time between the offset of the NS and the onset of the US.

Simultaneous Conditioning

- The onset of the NS and the onset of the US are simultaneous.
- Example:
 - *a tone and a shock are turned on at the same time*
- This procedure usually results in poor conditioning because the NS is no longer a good predictor of the US when they happen simultaneously.



Backward Conditioning

- The US is presented first and the NS is presented later.
- Example:
 - *the rat receives a shock and then hears a tone*
- This is the least effective procedure for conditioning.



Backward Excitatory Conditioning

- Backward excitatory conditioning can be achieved when the NS is a “biologically relevant” stimulus for fear (Keith-Lucas & Guttman, 1975).
- Many animals have an inherited predisposition to fear certain types of events (Seligman, 1971).
- Example:
 - *Rats have an inherited predisposition to fear snakes.*
 - *Poisonous snakes have constituted a significant threat to rats.*
 - *Even if the snake is presented after the shock, the fear elicited by the shock still becomes associated with the snake.*

Backward Inhibitory Conditioning

- Example:
 - *A tone sounds just as a shock is being terminated.*
 - *The tone essentially predicts the removal of shock.*
 - *The tone in this case may become a safety signal (CS–) that inhibits the occurrence of fear.*
 - *Bullying?*
 - *Presence of teacher*

Pseudoconditioning

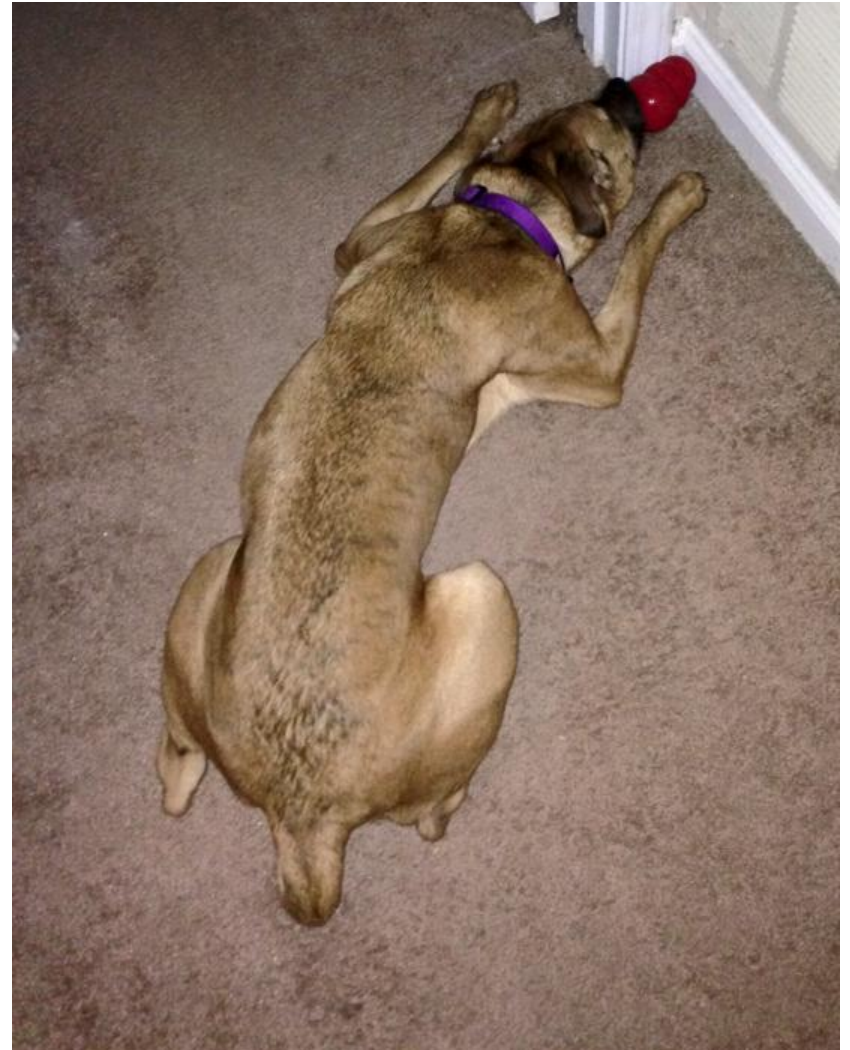
- Light flash: Shock → *Leg flexion*
 - Conditioning trials
- Light flash → *Leg flexion*
 - This looks like classical conditioning
- But....
- Beep → *Leg flexion*
 - This is not the CS

Pseudoconditioning

- The CR is elicited in the presence of an unassociated NS
 - Stimulus generalization

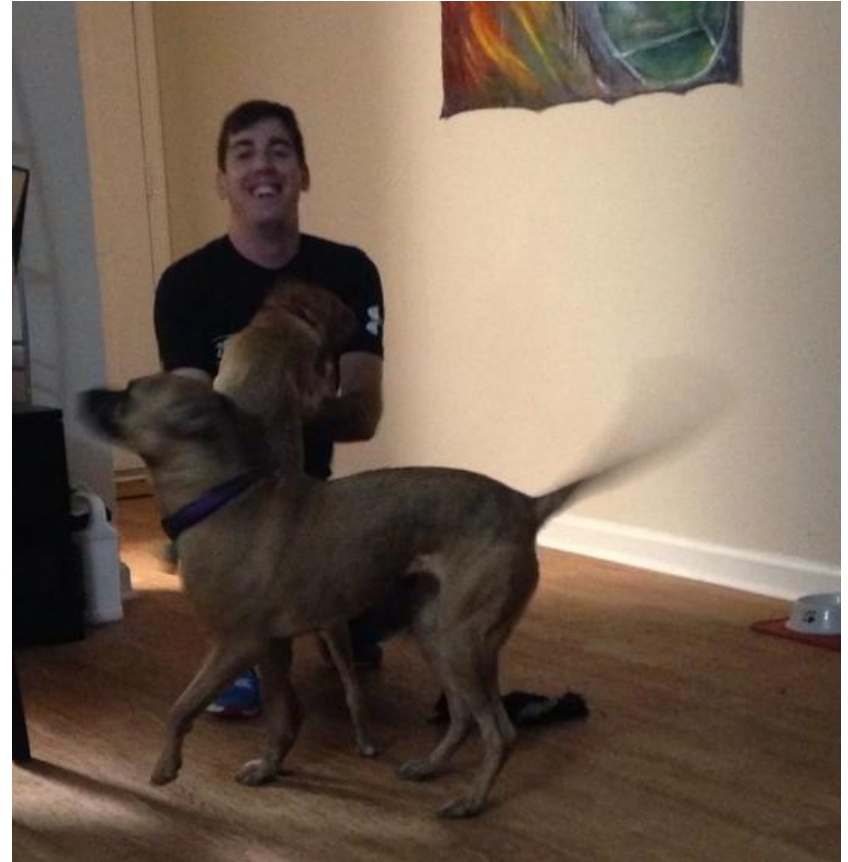
Pseudoconditioning

- Shelter dogs?



Pseudoconditioning

- Shelter dogs?



Summary

- Elicited behaviors are involuntary reactions to specific stimuli.
- Examples:
 - *reflexes*
 - *fixed action patterns*
- Repeated presentations of the same stimulus may:
 - decrease the strength of a behavior (*habituation*) or
 - increase the strength of a behavior (*sensitization*)depending on the intensity and evolutionary significance of the eliciting stimulus.

Summary, continued...

- In classical conditioning, a neutral stimulus is associated with some other stimulus that naturally elicits a response, and as a result the neutral stimulus also comes to elicit a response.
- In appetitive conditioning, the US is considered pleasant and something that an organism seeks out such as food.
- In aversive conditioning, the US is considered unpleasant and that an organism usually avoids such as a shock.

Summary, continued...

- The unconditioned stimulus (US) is the stimulus that naturally elicits a response, and the unconditioned
- response (UR) is the response that is naturally elicited by the US.
- The conditioned stimulus (CS) is the stimulus that, although initially a neutral stimulus (NS), comes to elicit a response because it has been associated
- with the US.
- The conditioned response (CR) is the response that is elicited by the CS.

Summary, continued...

- There are various temporal arrangements of the NS and US in classical conditioning:
 - Delayed Conditioning.
 - Trace Conditioning.
 - Simultaneous Conditioning.
 - Backward Conditioning.